

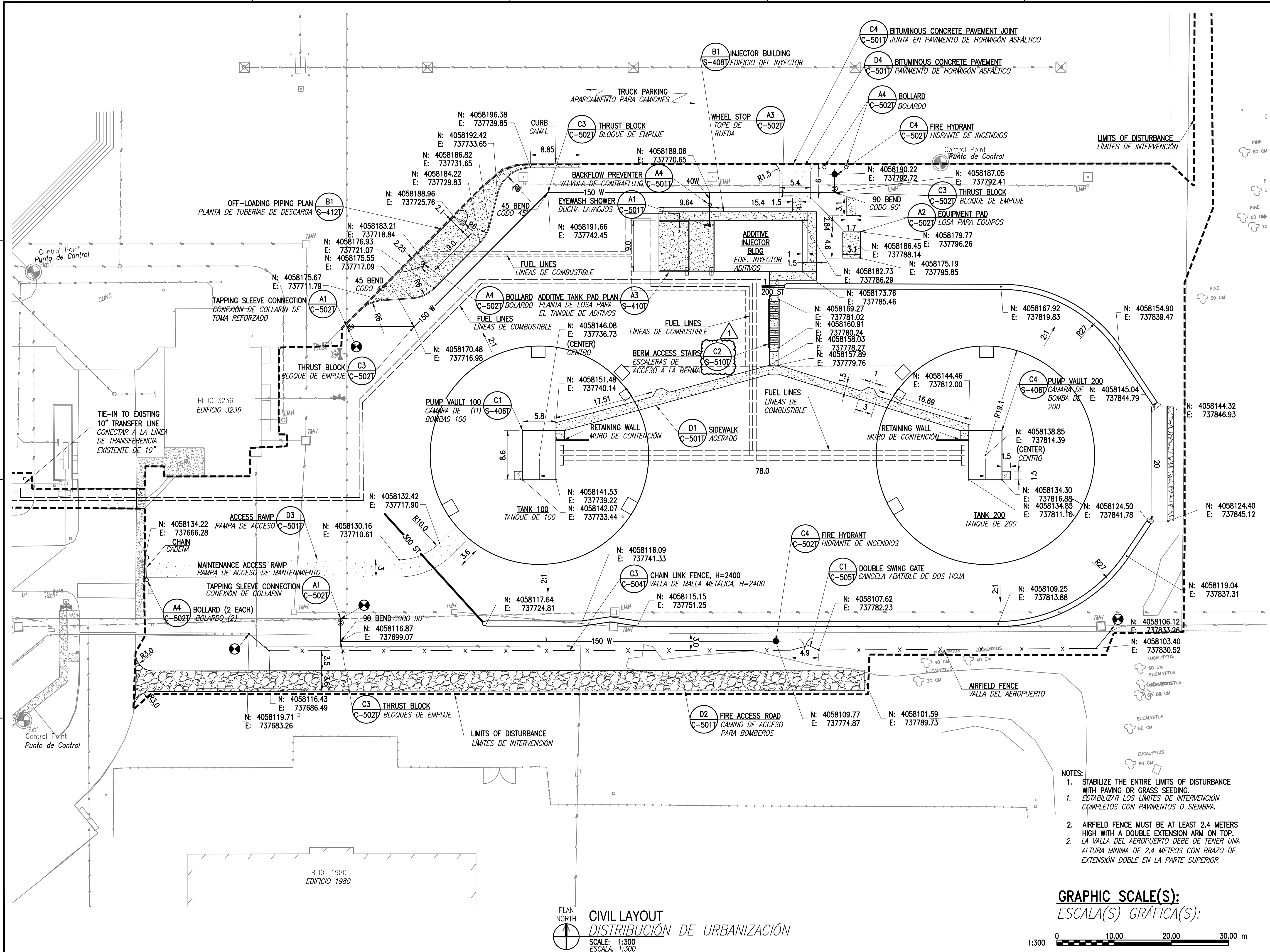
FILE NAME: P:\MBA\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Rota\CA0\CA0-C-101T CIVL LAYOUT.dwg LAYOUT NAME: C-101T PLOTTED: Thursday, January 29, 2026 - 9:42am USER: htopkins

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DATE		01/30/26			
SYN		DESCRIPTION			
1		CORRECT DETAIL CALLOUT			
APPROVED					
FOR COMMANDER NAVFAC					
ACTIVITY					
SATISFACTORY TO		DATE			
DES	RGB	DRW	RTH	CHK	JDP
FM/SM	CDA/ELA				
BRANCH MANAGER					
DESIGN MANAGER					
FIRE PROTECTION					
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND		NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC			
ATLANTIC DESIGN AND CONSTRUCTION		NORFOLK, VA			
NAVAL STATION ROTA		ROTA, SPAIN			
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1		CIVIL LAYOUT			
DISTRIBUCIÓN DE URBANIZACIÓN					
SCALE:					
EPROJECT NO.:					
CONSTR. CONTR. NO.:					
NAVFAC DRAWING NO.:		14140117			
SHEET		13 OF			
C-101T					
NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018					

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FILE NAME: P:\MBA\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Roto\CAO\WM-001T.dwg LAYOUT NAME: m-001 PLOTTED: Thursday, January 29, 2026 - 9:42am USER: hopkins

MECHANICAL ABBREVIATIONS AND SYMBOLS

ABREVIATURAS Y SÍMBOLOS DE MECÁNICA

AE	AIR ELIMINATOR VÁLVULA DE CONTROL DE AIRE	LCV	LEVEL CONTROL VALVE VÁLVULA DE CONTROL DE NIVEL
ATG	AUTOMATIC TANK GAUGE SENSOR AUTOMÁTICO DEL TANQUE	LPD	LOW POINT DRAIN DRENAJE PUNTO BAJO
BLDG	BUILDING EDIFICIO	M	METER METRO
BPCV	BACK PRESSURE CONTROL VALVE VÁLVULA DE CONTROL DE CONTRAPRESIÓN	mm	MILLIMETER MILÍMETRO
CONC	CONCRETE OR CONCENTRIC HORMIGÓN O CONCÉNTRICO	MIN	MINIMUM MÍNIMO
CS	CARBON STEEL ACERO AL CARBONO	NO	NUMBER NÚMERO
CV	CHECK VALVE OR CONTROL VALVE VÁLVULA ANTIRRETORNO O VÁLVULA DE CONTROL	OW	OILY WASTE RESIDUOS ACEITOSOS
DBB	DOUBLE BLOCK AND BLEED DOBLE BLOQUEO Y PURGA	PI	PRESSURE INDICATOR INDICADOR DE PRESIÓN
DEMO	DEMOLITION OR DEMOLISH DEMOLICIÓN O DEMOLER	PRT	PRODUCT RECOVERY TANK TANQUE DE RECUPERACIÓN DE PRODUCTO
DRN	DRAIN DRENAJE	RED	REDUCER REDUCTOR
ECC	ECCENTRIC EXCÉNTRICO	SCH	SCHEDULE CUADRO, SCHEDULE
FCV	FLOW CONTROL VALVE VÁLVULA DE CONTROL DE CAUDAL	SDD	SAFETY DEVICE AGAINST DETONATION, DRY TYPE DISPOSITIVO DE SEGURIDAD CONTRA DETONACIÓN, TIPO SECO
FOB	FLAT ON BOTTOM PLANO SOBRE BASE	SDL	SAFETY DEVICE AGAINST DETONATION, LIQUID TYPE DISPOSITIVO DE SEGURIDAD CONTRA DETONACIÓN, TIPO LÍQUIDO
FL	FUSIBLE LINK ENLACE FUSIBLE	SFI	SIGHT FLOW INDICATOR INDICADOR VISUAL DE CAUDAL
FPBV	FULL PORT BALL VALVE VÁLVULA DE BOLA DE PUERTO COMPLETO	SP	STRIPPING PUMP BOMBA DE EXTRACCIÓN
FSCV	FILTER SEPARATOR CONTROL VALVE VÁLVULA DE CONTROL DEL S EPARADOR DE FILTRO	SPL	SPADE PLATE/SPECTICLE BLIND
FQI	FLOW QUANTITY INDICATOR INDICADOR DE CANTIDAD DE CAUDAL	SS	STAINLESS STEEL ACERO INOXIDABLE
FS	FLOW SWITCH INTERRUPTOR DE CAUDAL	STA	STATION ESTACIÓN
FSI	FILTER/SEPARATOR (ISSUE) FILTRO/SEPARADOR (EMISOR)	TRV	THERMAL RELIEF VALVE VÁLVULA DE ALIVIO TÉRMICO
FSP	FUEL SAMPLE POINT PUNTO DE MUESTRA DE COMBUSTIBLE	TYP	TYPICAL TÍPICO
FSII	FUEL SYSTEM ICING INHIBTOR INHIBIDOR DE HIELO DEL SISTEMA DE COMBUSTIBLE	UON	UNLESS OTHERWISE NOTED A MENOS QUE SE INDIQUE OTRA COSA
FSR	FILTER/SEPARATOR (RECEIPT) FILTRO/SEPARADOR (RECEPTOR)	W/	WITH CON
HP	HORSEPOWER OR HEAT PUMP CABALLO DE VAPOR O BOMBA DE CALOR	WNF	WELD NECK FLANGE DIÁMETRO
HPV	HIGH POINT VENT VENTEO PUNTO ALTO	Ø	DIAMETER BRIDA CON CUELLO DE SOLDADURA
IF	INSULATING FLANGE BRIDA AISLANTE	ℙ	PLATE PLETINA
IV	ISSUE VENTURI EMISOR VENTURI	%	PERCENT PORCENTAJE
JP5	JP-5 JET FUEL SURTIDOR DE COMBUSTIBLE JP-5	#	NUMBER NÚMERO
kPa	KILO PASCAL KILOPASCAL		
LA	LIQUID PROBE SONDA DE LÍQUIDOS		

MECHANICAL LEGEND

LEYENDA DE MECÁNICA

SYMBOL SÍMBOLO	DESCRIPTION DESCRIPCIÓN
	FLOW OR PRESSURE CONTROL VALVE VÁLVULA DE CONTROL DE CAUDAL O DE PRESIÓN
	DOUBLE BLOCK AND BLEED PLUG VALVE VÁLVULA DE DOBLE BLOQUEO Y PURGA
	FLOW SWITCH INTERRUPTOR DE CAUDAL
	THERMAL OR SAFETY RELIEF VALVE VÁLVULA DE ALIVIO TÉRMICO O DE SEGURIDAD
	BALL VALVE VÁLVULA DE BOLA
	BUTTERFLY VALVE VÁLVULA DE MARIPOSA
	WELD NECK FLANGE VÁLVULA CON CUELLO DE SOLDADURA
	CONCENTRIC REDUCER REDUCTOR CONCÉNTRICO
	PRESSURE INDICATOR INDICADOR DE PRESIÓN
	PUMP (TRIANGLE INDICATES FLOW DIRECTION) BOMBA (EL TRIÁNGULO DE LA BOMBA INDICA LA DIRECCIÓN DEL FLUJO)
	MALE CAM TYPE DISCONNECT WITH FEMALE DUST CAP DESCONECTOR MACHO TIPO LEVA CON GUARDAPOLVO HEMBRA
	CONNECT TO EXISTING CONECTAR AL EXISTENTE
	MOTOR ACTUATOR ACTUADOR DEL MOTOR
	AIR ELIMINATOR ELIMINADOR DE AIRE
	UNDERGROUND LINE LÍNEA ENTERRADA
	ABOVEGROUND LINE LÍNEA SOBRE EL TERRENO
	PIPE UTILITY LINE ABBREVIATION ABREVIADO DE LÍNEA DE TUBERÍA
	SIZE TAMAÑO
	MATERIAL TYPE TIPO DE MATERIAL
	BITUMINOUS CONCRETE (PLAN) HORMIGÓN BITUMINOSO (PLANTA)
	CONCRETE (PLAN) HORMIGÓN
	CONCRETE (SECTION) HORMIGÓN (SECCIÓN)

DEMOLITION NOTES

NOTAS DE DEMOLICIÓN

- EXISTING FUEL SYSTEM DEMOLITION IS VERY LIMITED FOR THIS PROJECT. THE ONLY FUEL DEMOLITION REQUIRED IS ON THE INCOMING 300 mm RECEIPT PIPELINE TO ACCOMMODATE THE NEW TIE-IN VALVE VAULT. ALL OTHER ASPECTS OF THIS PROJECT ARE NEW AND DO NOT REQUIRE DEMOLITION OF EXISTING FUEL SYSTEM COMPONENTS.
- UNLESS OTHERWISE INDICATED, ALL EXISTING FUEL PIPING IS EXTERIOR COATED CARBON STEEL.
- "DEMOLISH" - REMOVE IN ITS ENTIRETY.
- "ABANDON IN PLACE" - CLEAN PIPING BY PIGGING, FILL COMPLETELY WITH GROUT, AND CAP OPEN ENDS OF PIPE WITH A WELDED PIPE CAP OR BLIND FLANGE.
- THE ACTIVITY WILL, TO THE FULLEST EXTENT POSSIBLE, EMPTY THE ACTIVE UNDERGROUND AND ABOVEGROUND JP-5 PIPELINES, TANKS AND EQUIPMENT. HOWEVER, FOR BASIS OF BIDS THE PIPES ARE FULL OF FUEL ON TOP OF 50 mm OF PETROLEUM CONTAMINATED SLUDGE AND WATER, AND THE TANKS ARE FULL OF FUEL ON TOP OF 150 mm OF PETROLEUM CONTAMINATED SLUDGE AND WATER. THE CONTRACTOR MUST DRAIN, COLLECT, AND TURN OVER TO THE GOVERNMENT ANY FUEL NOT EMPITIED BY THE ACTIVITY.

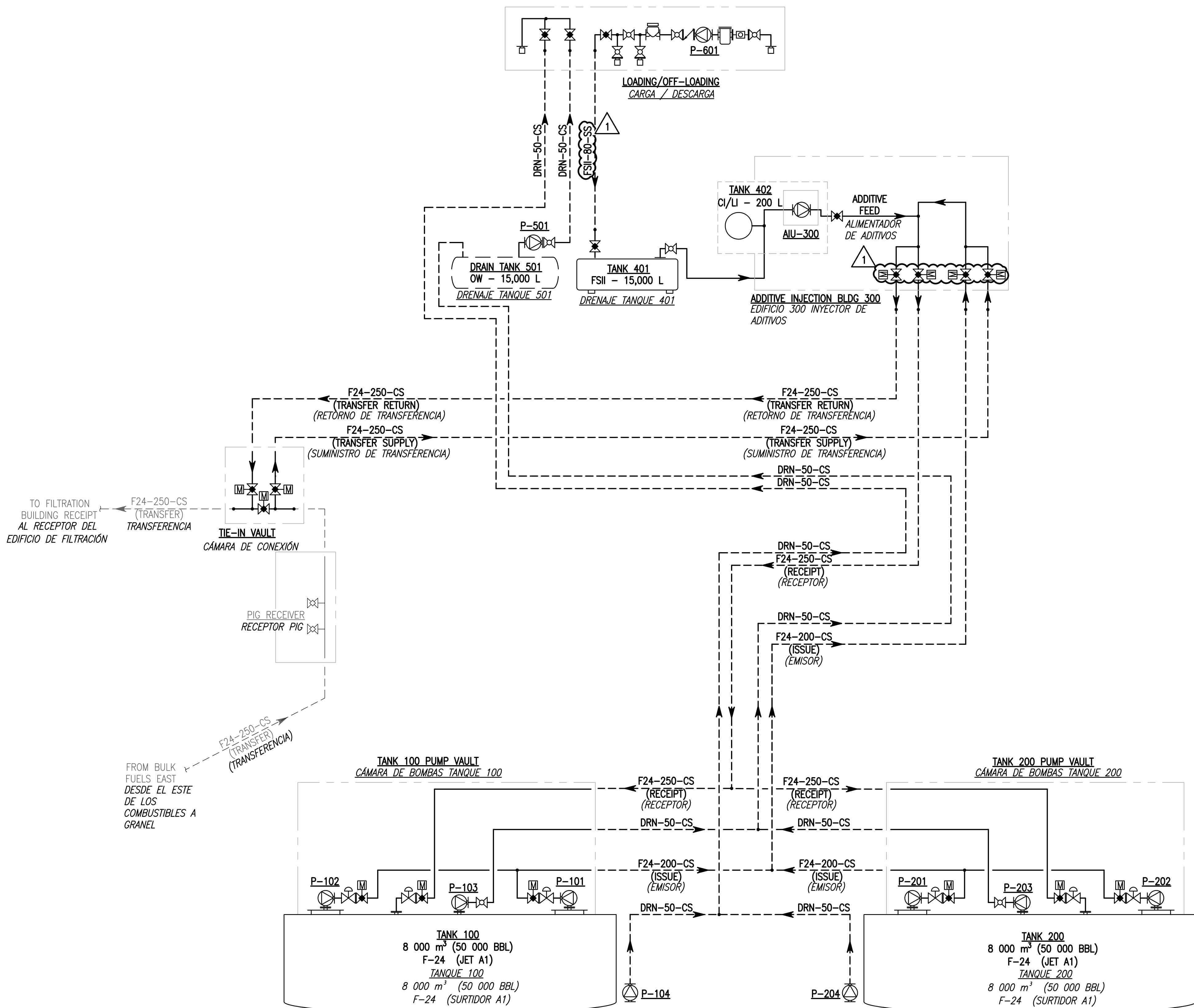
- ALL PIPING IS EXTERIOR COATED CARBON STEEL, BOTH ABOVEGROUND AND UNDERGROUND AS INDICATED, WITH THE EXCEPTION OF THE PIPING AND EQUIPMENT FOR THE FUEL ADDITIVE SYSTEM OR AS OTHERWISE NOTED. FUEL ADDITIVE SYSTEM PIPING AND EQUIPMENT IS STAINLESS STEEL, BOTH ABOVEGROUND AND UNDERGROUND AS INDICATED.
- PROVIDE HIGH POINT VENTS WHERE INDICATED AND AT ALL HIGH POINTS. PROVIDE LOW POINT DRAINS WHERE INDICATED AND AT ALL LOW POINTS. INSTALL FUEL PIPING WITH A MINIMUM SLOPE OF 0.2% DOWN FROM HIGH POINTS TO LOW POINTS, UNLESS OTHERWISE INDICATED.
- CONTRACTOR MUST PERFORM EXPLORATORY OPERATIONS AS NECESSARY TO VERIFY THE LOCATIONS, ELEVATIONS, AND DIMENSIONS OF ALL EXISTING UNDERGROUND UTILITIES AND OBSTRUCTIONS. HAND EXCAVATE WITHIN 1 METER OF EXISTING UTILITIES.
- EXISTING FUEL SYSTEMS MUST REMAIN IN OPERATION THROUGHOUT THE PROJECT EXCEPT FOR INDIVIDUAL PIPELINES AND SYSTEMS TAKEN OUT OF SERVICE TO MAKE THE FINAL PIPING CONNECTIONS.
- BURIED FUEL PIPING MUST NOT BE BACK FILLED WITHOUT COMPLETING THE COATING, PNEUMATIC, WELD RADIOGRAPHY AND OTHER PIPING EXAMINATIONS AND APPROVALS. IF INSTALLED PIPE IS EMPTY WITHOUT PRODUCT FOR MORE THAN 30 DAYS, CONTRACTOR MUST CHARGE INACTIVE PIPELINES WITH A NITROGEN BLANKET TO PREVENT INTERNAL PIPE CORROSION. ENERGIZE AND BALANCE THE CURRENT CATHODIC PROTECTION SYSTEMS WITHIN 30 DAYS OF BACK FILLING ANY PIPING.
- THIS MILCON PROJECT FEATURES TWO DISTINCTLY SEPARATE FUEL SYSTEM COMPONENTS, THE F-24 STORAGE TANK AREA AND THE JP-5 CARGO PIPELINE. THERE IS NO INTERCONNECTION BETWEEN THE TWO PROJECTS, AND PIPING, EQUIPMENT, AND COMPONENTS BETWEEN THE TWO SEPARATE SYSTEMS MAY BE DIFFERENT PIPING MATERIALS, DIFFERENT COATING SYSTEMS, DIFFERENT PRESSURE CLASS, ETC. CONTRACTOR MUST BE AWARE THAT DETAILS SHOWN ON VARIOUS SHEETS AND ITEMS LISTED WITHIN THE SPECIFICATIONS MAY APPLY TO THE F-24 STORAGE TANK SYSTEM, THE JP-5 PIPELINE SYSTEM, OR BOTH (AS INDICATED OR NOTED).
- HPV, LPD, TRV, PI, FSP, AND IF ARE INDICATED ON THE PLANS WITH A LEADER AND AN ABBREVIATION. THE DETAILS ARE LOCATED AS FOLLOWS:

- SDD'S AND SDL'S MUST BE THE SAME SIZE AS THE CONNECTED PIPING.



DATE	01/30/26
SYN	1
DESCRIPTION	REVISED NOTES
SEAL	
A/E INFO	
FOR COMMANDER NAVFAC	
ACTIVITY	
SATISFACTORY TO	DATE
DES	BMS
DRAW	RTH
CHK	DWN
PM/CM	CC/VEA
BRANCH MANAGER	
DESIGN MANAGER	
FIRE PROTECTION	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND	ATLANTIC
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND	ATLANTIC
NAVAL STATION	ROTA, SPAIN
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1	
MECHANICAL ABBREVIATIONS	ABREVIATURAS DE MECÁNICA
SCALE:	
PROJECT NO.:	
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	14140177
SHEET	73 OF 342
M-001T	

FILE NAME: P:\MBA\21-Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Foto\CAO\WM-0027 CUT-AND-COVER FUEL SYSTEM FLOW SCHEMATIC.dwg LAYOUT NAME: m-002 PLOTTED: Monday, February 23, 2026 - 8:49am USER: hopkins



CUT-AND-COVER FUEL SYSTEM FLOW SCHEMATIC
DIAGRAMA DE FLUJO DEL SISTEMA DE COMBUSTIBLE
SCALE: NONE
ESCALA: SIN ESCALA

FUELING SYSTEM NOTES NOTAS DEL SISTEMA DE COMBUSTIBLE

1. EACH OF THE NEW BULK STORAGE TANKS (TANK 100 AND TANK 200) WILL BE PROVIDED WITH A PUMP VAULT AND (2) 600 GPM VERTICAL TURBINE PUMPS. MAXIMUM FLOW RATE WITH BOTH PUMPS RUNNING SIMULTANEOUSLY WILL BE 1,200 GPM.
1. CADA UNO DE LOS NUEVOS TANQUES DE ALMACENAMIENTO A GRANEL (TANQUE 100 Y TANQUE 200) SE PROVEERÁ CON UNA CÁMARA DE BOMBAS Y (2) BOMBAS VERTICALES DE TURBINA DE 600 GPM. EL CAUDAL MÁXIMO SERÁ DE 1.200 GPM CON AMBAS BOMBAS FUNCIONANDO SIMULTÁNEAMENTE.
2. THE ADDITIVE INJECTION BUILDING (BLDG 300) HAS THE CAPABILITY TO ALLOW FUEL TO PASS THROUGH OR TO ADDITIVE FUEL TO ACCOMMODATE SEVERAL DIFFERENT TRANSFER SCENARIOS. ADDITIVE INJECTION SKID CAN BE ISOLATED FROM PIPING SYSTEM USING ISOLATION VALVES IN THE FEED LINE.
2. EL EDIFICIO DEL INYECTOR DE ADITIVOS (EDIFICIO 300) TIENE CAPACIDAD PARA PERMITIR EL PASO DEL COMBUSTIBLE A TRAVÉS DE ÉL O PARA ADITIVARLO CON EL FIN DE ADAPTARSE A VARIOS ESCENARIOS DE TRANSFERENCIA DIFERENTES. EL SISTEMA DE INYECCIÓN DE ADITIVOS PUEDE AISLARSE DEL SISTEMA DE TUBERÍAS MEDIANTE VÁLVULAS DE AISLAMIENTO EN LA LÍNEA DE ALIMENTACIÓN.
3. FOR ANY OF THE ADDITIZATION SCENARIOS, THE FLOW RATE WILL BE 1,200 GPM. THE VARIOUS ADDITIZATION TRANSFER SCENARIOS ARE AS FOLLOWS:
3. PARA CUALQUIERA DE LOS ESCENARIOS DE ADICIÓN, EL CAUDAL SERÁ DE 1.200 GPM. LOS DISTINTOS ESCENARIOS DE TRANSFERENCIA DE ADICIÓN SON LOS SIGUIENTES:
 - FUEL VIA TRANSFER LINE IS ADDITIZED THEN RECEIVED TO EITHER OF THE NEW BULK STORAGE TANKS (TANK 100 AND TANK 200).
 - EL COMBUSTIBLE A TRAVÉS DE LA LÍNEA DE TRANSFERENCIA ES ADITIVADO Y ENTONCES RECIBIDO POR CUALQUIERA DE LOS NUEVOS TANQUES DE ALMACENAJE (TANQUE 100 Y TANQUE 200).
 - FUEL VIA TRANSFER LINE IS ADDITIZED THEN RECEIVED TO EITHER OF THE EXISTING HYDRANT SYSTEM OPERATING TANKS.
 - EL COMBUSTIBLE A TRAVÉS DE LA LÍNEA DE TRANSFERENCIA ES ADITIVADO Y ENTONCES RECIBIDO POR CUALQUIERA DE LOS TANQUES EN FUNCIONAMIENTO DEL SISTEMA DE HIDRANTES EXISTENTE.
 - FUEL WITHDRAWN FROM EITHER OF THE NEW BULK STORAGE TANK IS ADDITIZED THEN RECEIVED INTO EITHER OF THE EXISTING HYDRANT SYSTEM OPERATING TANKS.
 - EL COMBUSTIBLE EXTRAÍDO DE CUALQUIERA DE LOS NUEVOS TANQUES DE ALMACENAMIENTO SE ADITIVA Y LUEGO SE RECIBE EN CUALQUIERA DE LOS TANQUES EN FUNCIONAMIENTO DEL SISTEMA DE HIDRANTES EXISTENTE.
 - FUEL CAN BE WITHDRAWN FROM EITHER OF THE NEW BULK STORAGE TANKS, ADDITIZED, THEN RECEIVED INTO THE OTHER NEW BULK STORAGE TANK.
 - EL COMBUSTIBLE PUEDE RETIRARSE DE CUALQUIERA DE LOS NUEVOS TANQUES DE ALMACENAMIENTO A GRANEL, ADITIVARSE Y, A CONTINUACIÓN, RECIBIRSE EN EL OTRO NUEVO TANQUE DE ALMACENAMIENTO A GRANEL.



APPROVED	A/E INFO
FOR COMMANDER NAVFAC	ACTIVITY
SATISFACTORY TO: DATE	
DESIGNER: BMS	DRAWN: RTH
IN/OUT: CCA/ELA	CHK: DWN
BRANCH MANAGER	
DESIGN MANAGER	
FIRE PROTECTION	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND	COMMAND
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC	NORFOLK, VA
NAVAL STATION ROTA	ROTA, SPAIN
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1	
CUT-AND-COVER FUEL SYSTEM FLOW SCHEMATIC	
DIAGRAMA DE FLUJO DEL SISTEMA DE COMBUSTIBLE	
SCALE:	
PROJECT NO.:	
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	14140178
SHEET	74 OF 342
M-002T	

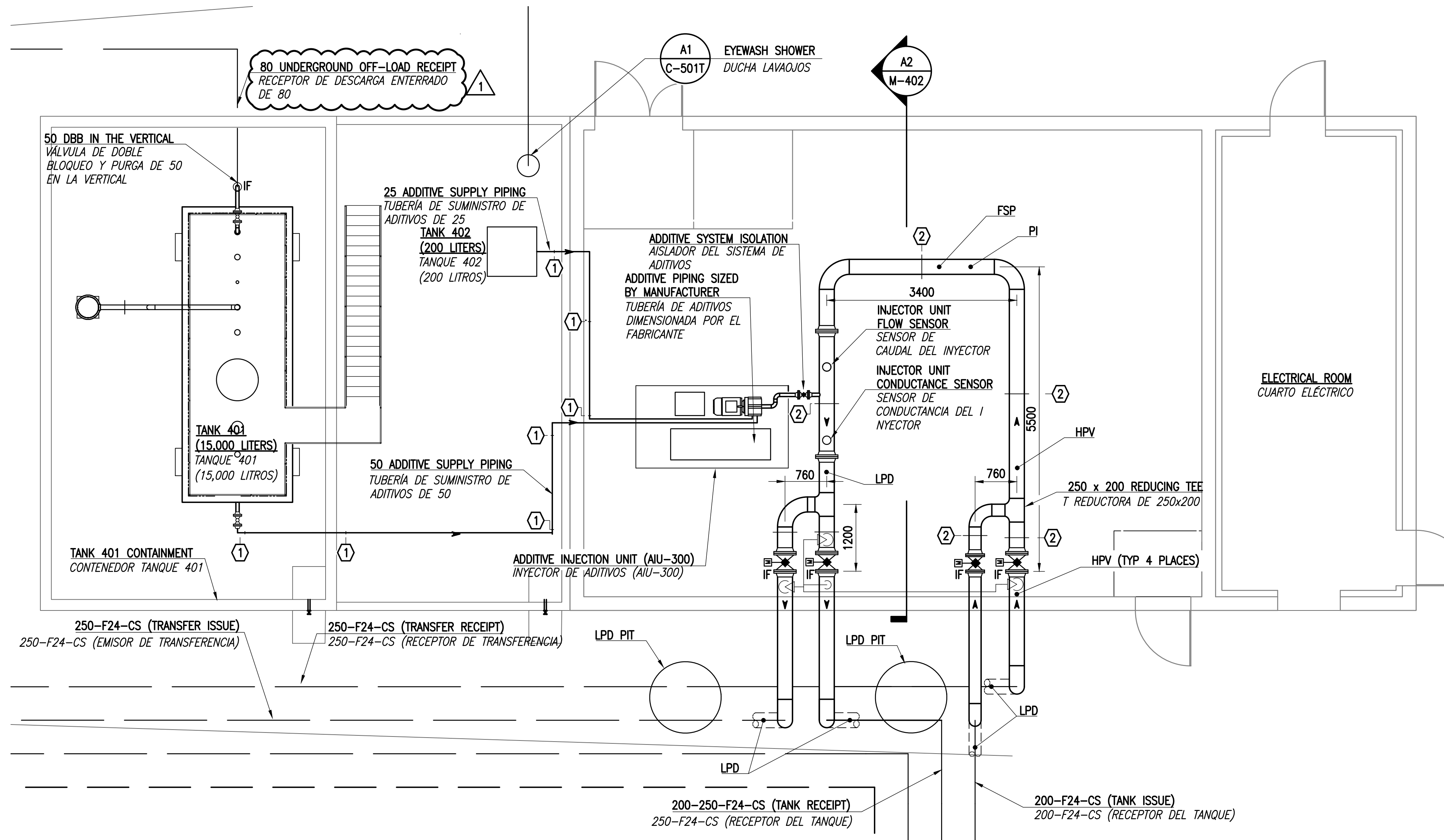
FILE NAME: P:\ABA\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Rota\CAO\WM-402 ADDITIVE INJECTOR BUILDING.dwg LAYOUT NAME: m-402 PLOTTED: Thursday, January 29, 2026 - 11:32am USER: rhopkins

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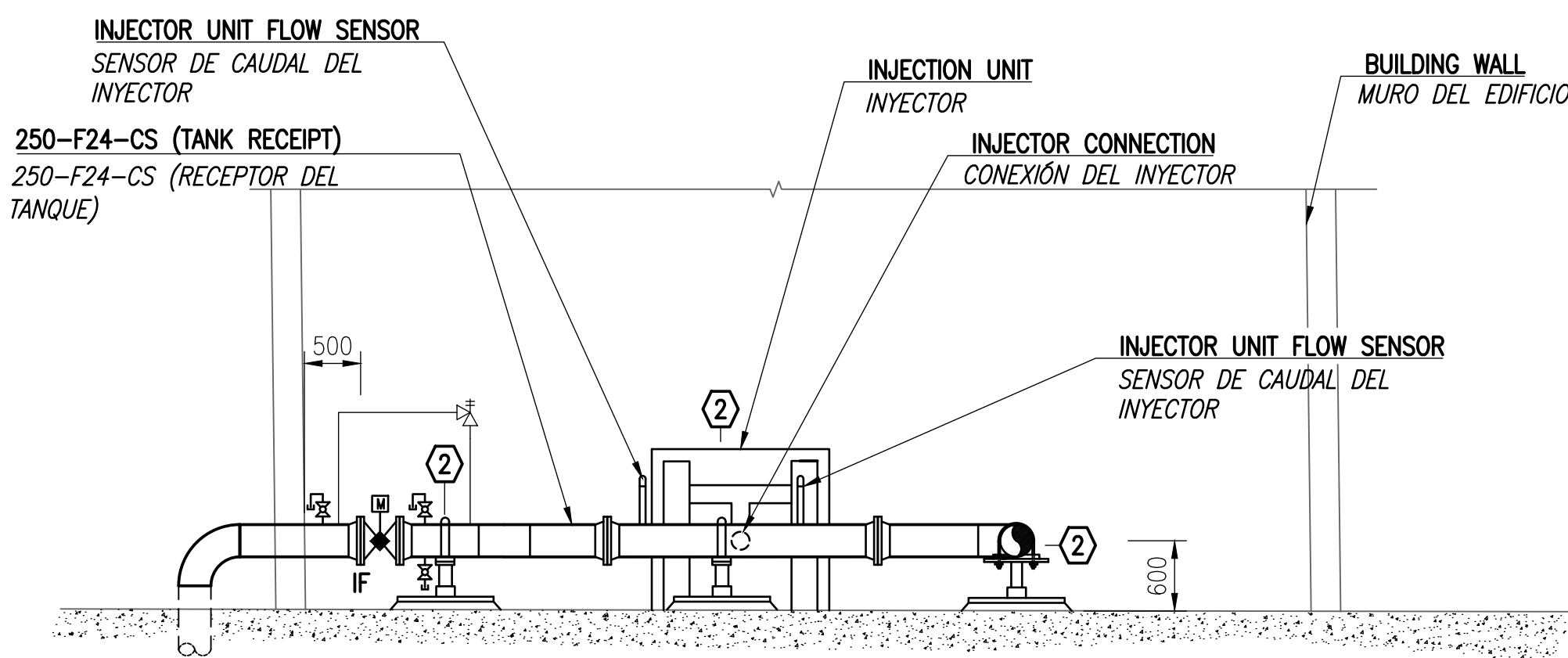
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ADDITIVE INJECTOR BUILDING PIPING PLAN
PLANTA DE TUBERÍAS DEL EDIFICIO DEL INYECTOR DE ADITIVOS
SCALE: 1:50
ESCALA: 1:50



SECTION
SECCIÓN
A2
SCALE: 1:50
ESCALA: 1:50

ADDITIVE INJECTOR BUILDING NOTES: NOTAS DEL EDIFICIO DEL INYECTOR DE ADITIVOS:

1. THE ADDITIVE INJECTION EQUIPMENT, CONTROLS, AND ACCESSORIES, INCLUDING THE SMALL CI/LI ADDITIVE TANK, MUST BE PROVIDED AS A SINGLE SYSTEM UNIT FROM ONE SUPPLIER. THE FSII TANK WILL BE PROVIDED AS A STAND ALONE TANK, BUT WILL BE CONNECTED TO THE INJECTION SKID FOR PIPING AND CONTROLS. FSII TANK SUPPLIER WILL PROVIDE AN ADEQUATELY SIZED DESICCANT DRYER.
EL EQUIPO DE INYECCIÓN DE ADITIVOS, CONTROLES Y ACCESORIOS, INCLUYENDO LOS DOS TANQUES DE ADITIVOS PEQUEÑOS, DEBERÁN SER SUMINISTRADOS EN UN ÚNICO SISTEMA DESDE UN PROVEEDOR. EL TANQUE FSII SERÁ SUMINISTRADO COMO UN DEPÓSITO INDEPENDIENTE PERO ESTARÁ CONECTADO A LA PLATAFORMA DE INYECCIÓN PARA LAS TUBERÍAS Y CONTROLES.
2. ADDITIVE INJECTION SYSTEM WILL INCLUDE INJECTION PUMP SYSTEM WITH ALL ASSOCIATED PIPING AND CONTROLS. THIS SYSTEM WILL HAVE THE CAPACITY TO INJECT (3) DIFFERENT ADDITIVES INTO THE FUEL STREAM. THIS SYSTEM WILL ADDITIZE THE F-24 WITH CI/LI AND FSII. THE FINAL ADDITIVE INJECTION SLOT WILL BE UNUSED BUT AVAILABLE FOR FUTURE USE IF THE NEED REQUIRES.
EL SISTEMA DE INYECCIÓN DE ADITIVOS INCLUIRÁ EL SISTEMA DE BOMBA DE INYECCIÓN, Y TODAS LAS TUBERÍAS Y CONTROLES ASOCIADOS Y TENDRÁ CAPACIDAD PARA INYECTAR (3) ADITIVOS DIFERENTES EN EL CAUDAL DE COMBUSTIBLE. EL ADITIVO CI REQUERIRÁ UN DEPÓSITO DE ALMACENAMIENTO DEDICADO QUE SE PROVEERÁ COMPLETO COMO PARTE DEL SISTEMA DE INYECCIÓN DE ADITIVOS. EL ADITIVO FSII SERÁ ALMACENADO EN UN TANQUE AÉREO CERCANO Y SEPARADO.



A/E INFO

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO: DATE

DES: BMS | DRW: RTH | CHK: DWN

PM/CM: CCA/ELA

BRANCH MANAGER

DESIGN MANAGER

FIRE PROTECTION

NAVAL FACILITIES ENGINEERING SYSTEM COMMAND

NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC

ATLANTIC DESIGN AND CONSTRUCTION

NAVAL STATION ROTA

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

ROTA, SPAIN

NORFOLK VA

ADDITIVE INJECTOR BUILDING PIPING PLAN

PLANTA DE TUBERÍAS DEL EDIFICIO DEL INYECTOR DE ADITIVOS

SCALE:

PROJECT NO.:

CONSTR. CONTR. NO.:

NAVFAC DRAWING NO.:

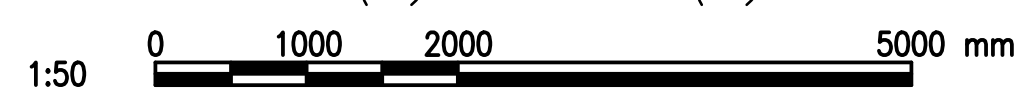
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SHEET 79 OF 342

M-402T

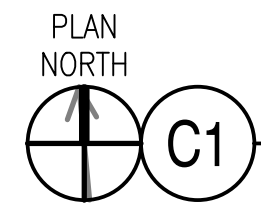
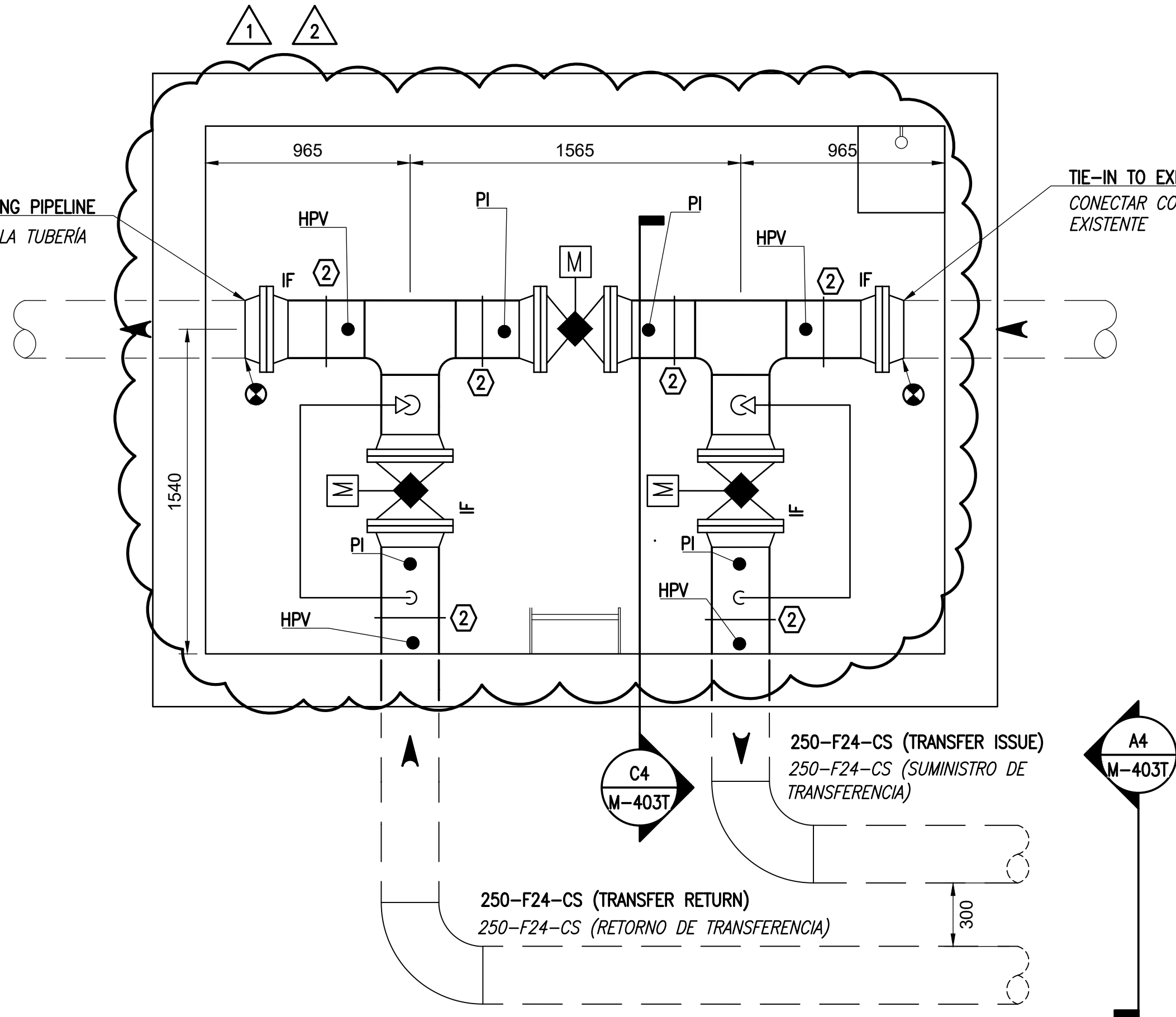
NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018

GRAPHIC SCALE(S):
ESCALA(S) GRÁFICA(S):



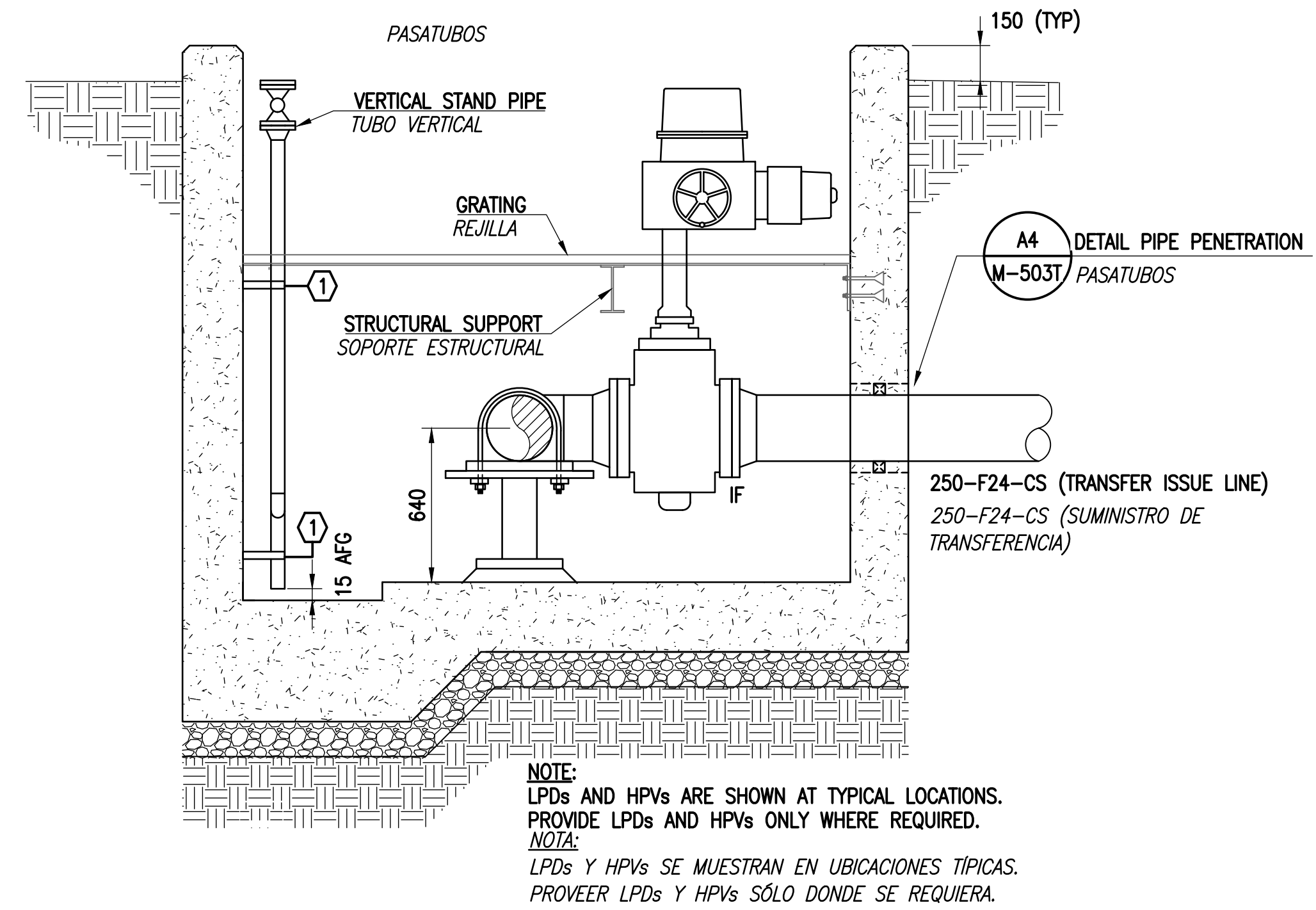
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TIE-IN TO EXISTING PIPELINE
CONECTAR CON LA TUBERÍA
EXISTENTE

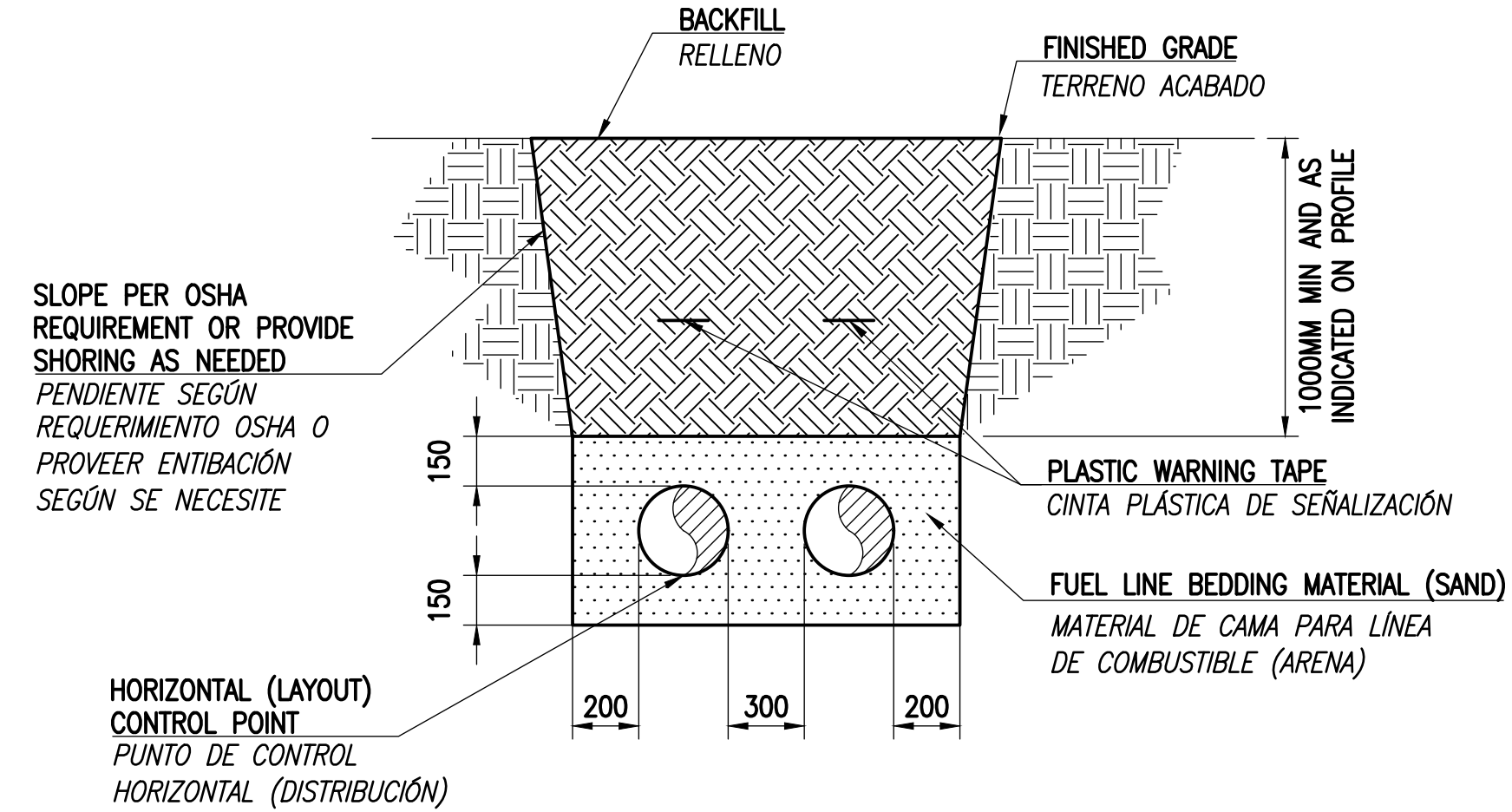


TIE-IN VAULT PIPING PLAN
PLANTA DE TUBERÍAS DE LA CÁMARA DE CONEXIÓN
SCALE: 1:20
ESCALA: 1:20

TIE-IN TO EXISTING PIPELINE
CONECTAR CON LA TUBERÍA
EXISTENTE



SECTION
SECCIÓN
C4
SCALE: 1:20
ESCALA: 1:20



SECTION
SECCIÓN
A4
SCALE: 1:20
ESCALA: 1:20

GRAPHIC SCALE(S):
ESCALA(S) GRÁFICA(S):

0 400 1000 2000 mm
1:20



APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DES BMS DRW RTH CHK DWN

PM/SM CCA/ELA

BRANCH MANAGER

DESIGN MANAGER

FIRE PROTECTION

NAVAL FACILITIES ENGINEERING SYSTEM COMMAND

NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC

ATLANTIC DESIGN AND CONSTRUCTION

NAVAL STATION ROTA

ROTA, SPAIN

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

TIE-IN VAULT PIT

FOSO DE LA CÁMARA DE CONEXIONES

SCALE:

PROJECT NO.:

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 14140184

SHEET 80 OF 342

M-403T

NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018

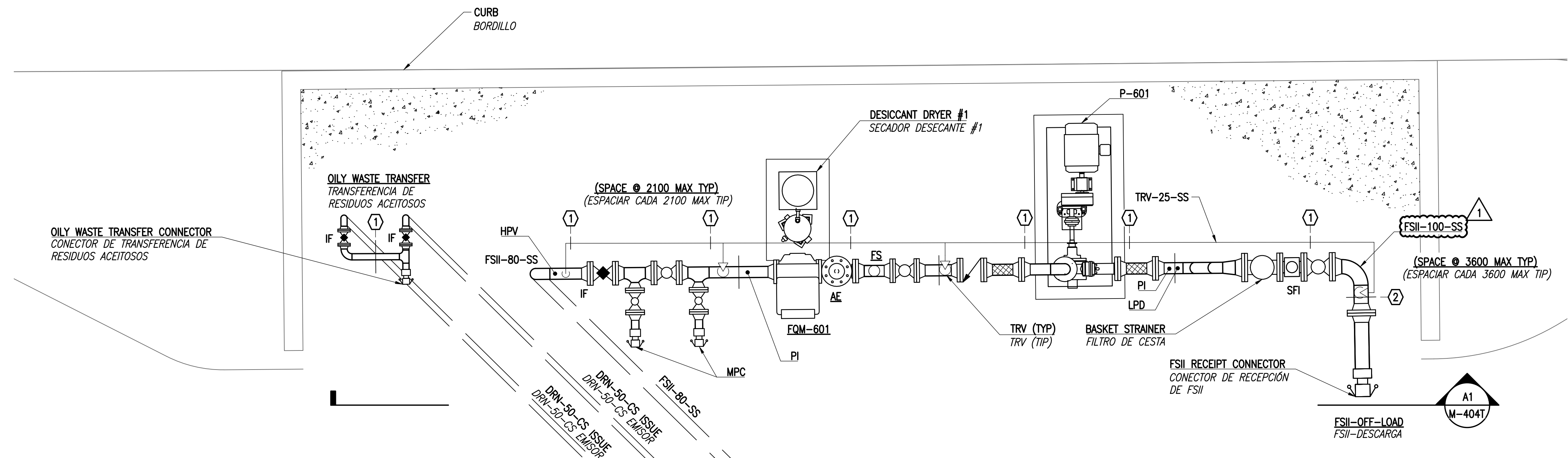
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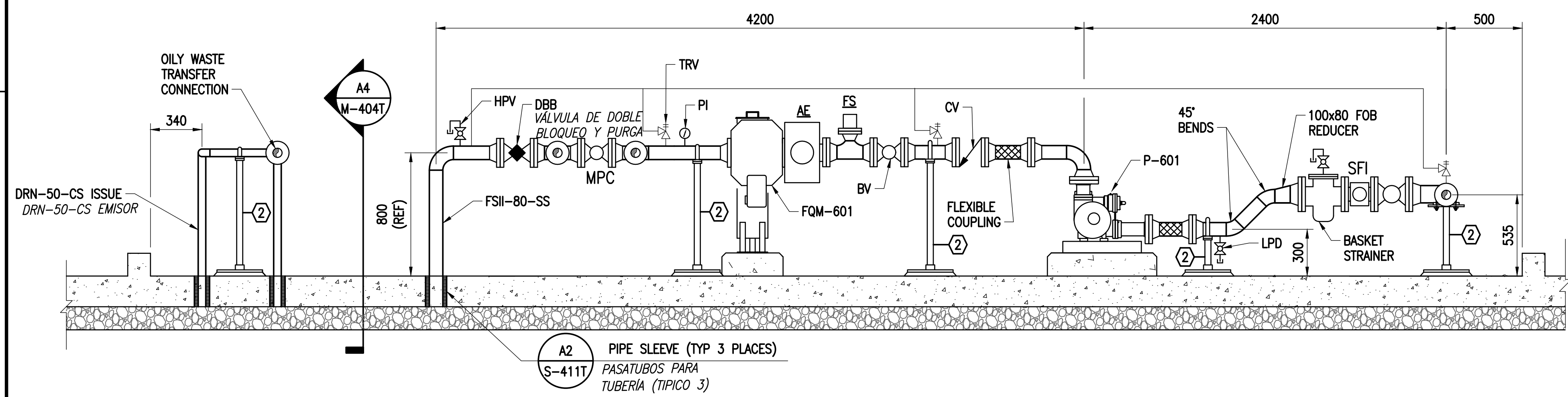
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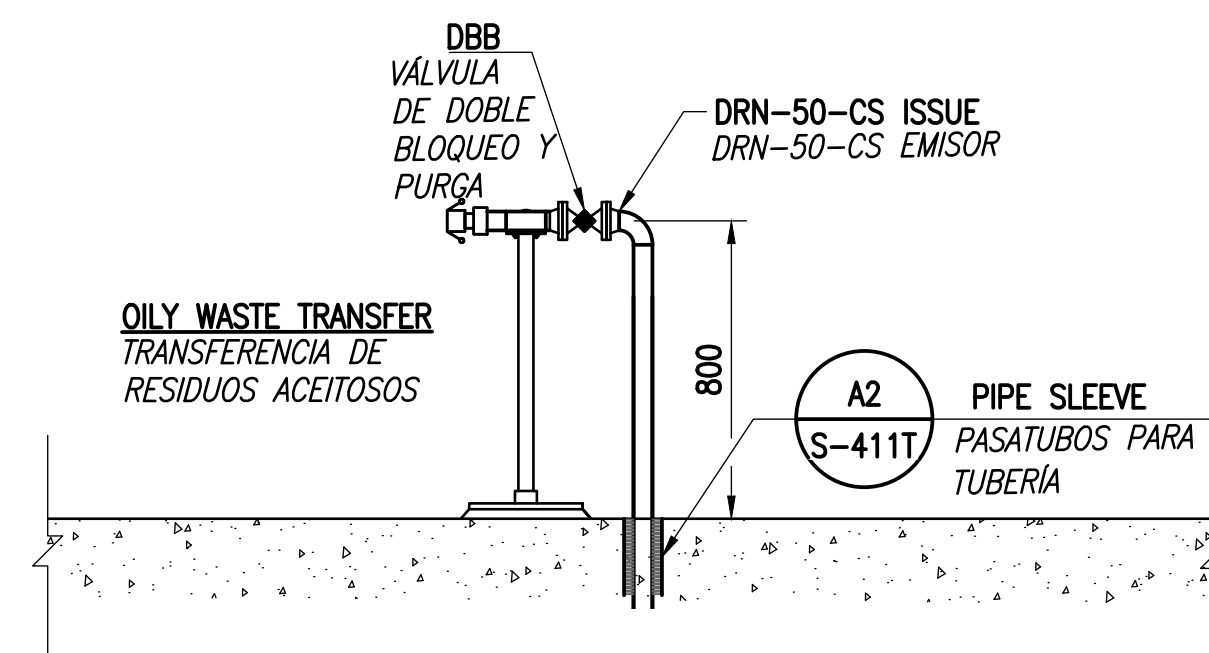
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PLAN NORTH
OFF-LOADING PIPING PLAN
PLANTA DE TUBERIAS DE DESCARGA
SCALE: 1:20
ESCALA: 1:20



A1
SECTION
SECCIÓN
SCALE: 1:20
ESCALA: 1:20



A4
SECTION
SECCIÓN
SCALE: 1:20
ESCALA: 1:20

GRAPHIC SCALE(S):
ESCALA(S) GRÁFICA(S):
0 400 1000 2000 mm
1:20



APPROVED		A/E, R/E	
FOR COMMANDER NAVFAC			
ACTIVITY			
Satisfactory to		DATE	
DESIGNER	BMS	DRAWN	RTH
PM/SM		CC/LIA	CHK
BRANCH MANAGER		DWN	
DESIGN MANAGER			
FIRE PROTECTION			
DEPARTMENT OF THE NAVY		NAVAL FACILITIES ENGINEERING SYSTEM COMMAND	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC		NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC	
ATLANTIC DESIGN AND CONSTRUCTION		ATLANTIC DESIGN AND CONSTRUCTION	
NAVAL STATION ROTA		ROTA, SPAIN	
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1			
		OFF LOADING	
		DESCARGA	
SCALE:			
PROJECT NO.:			
CONSTR. CONTR. NO.			
NAVFAC DRAWING NO.			
14140185			
SHEET			
81 OF 342			
M-404T			

NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018

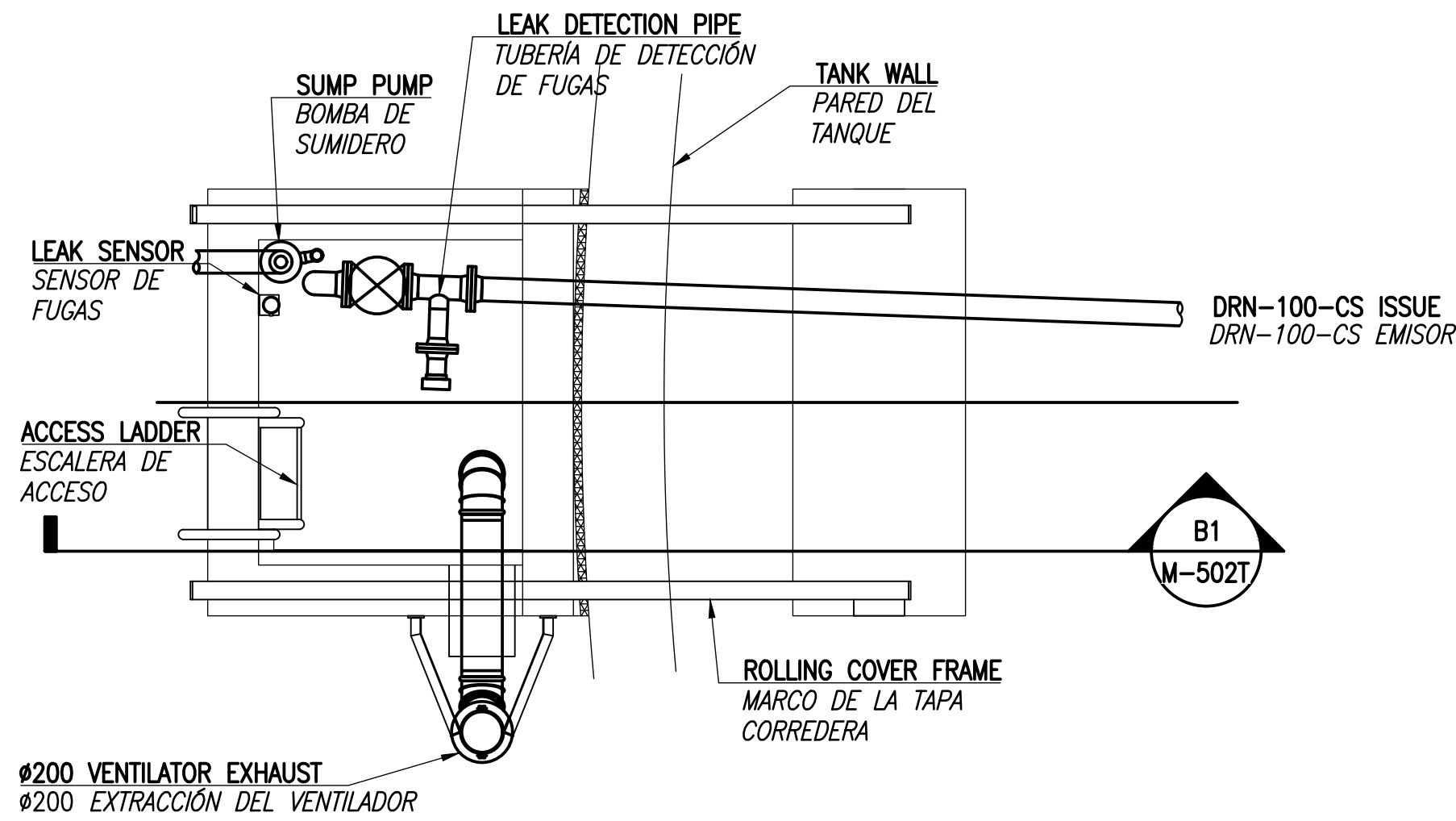
D

C

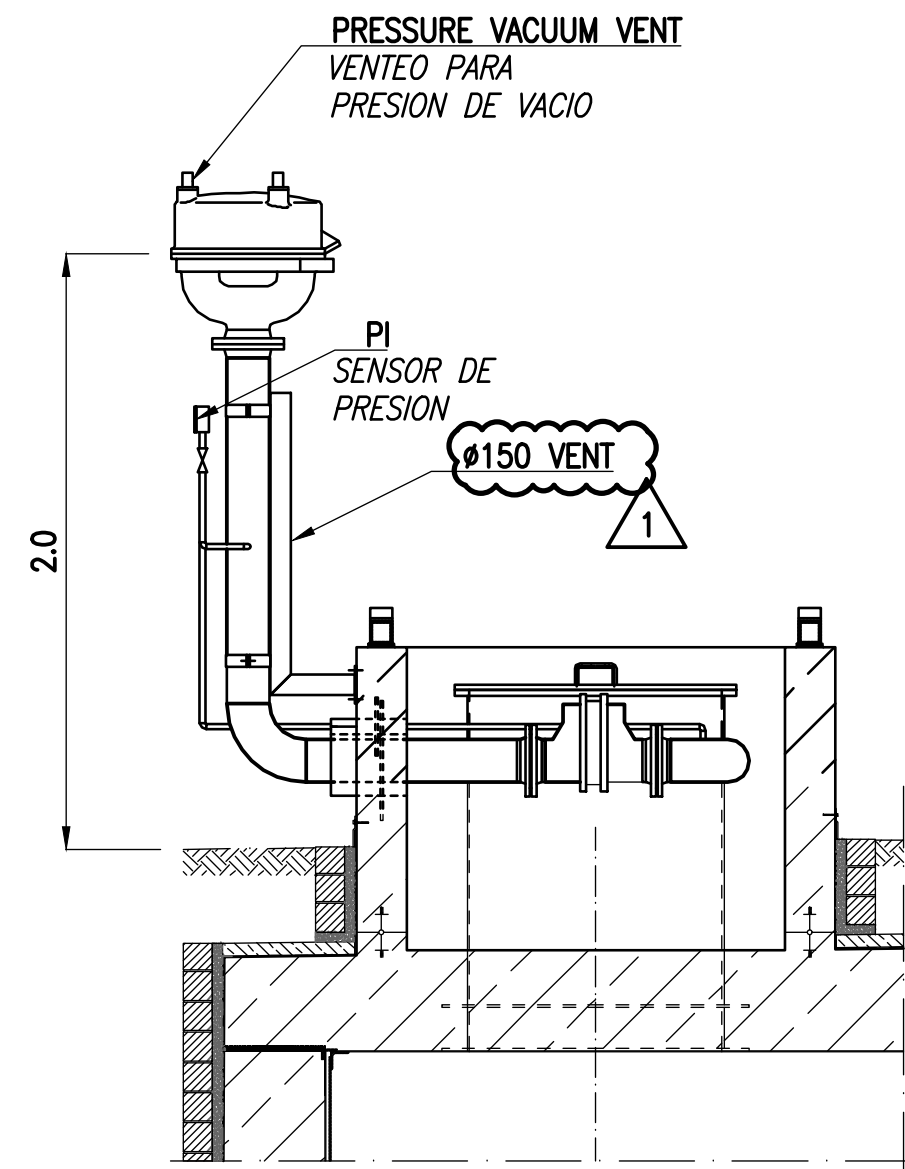
B

A

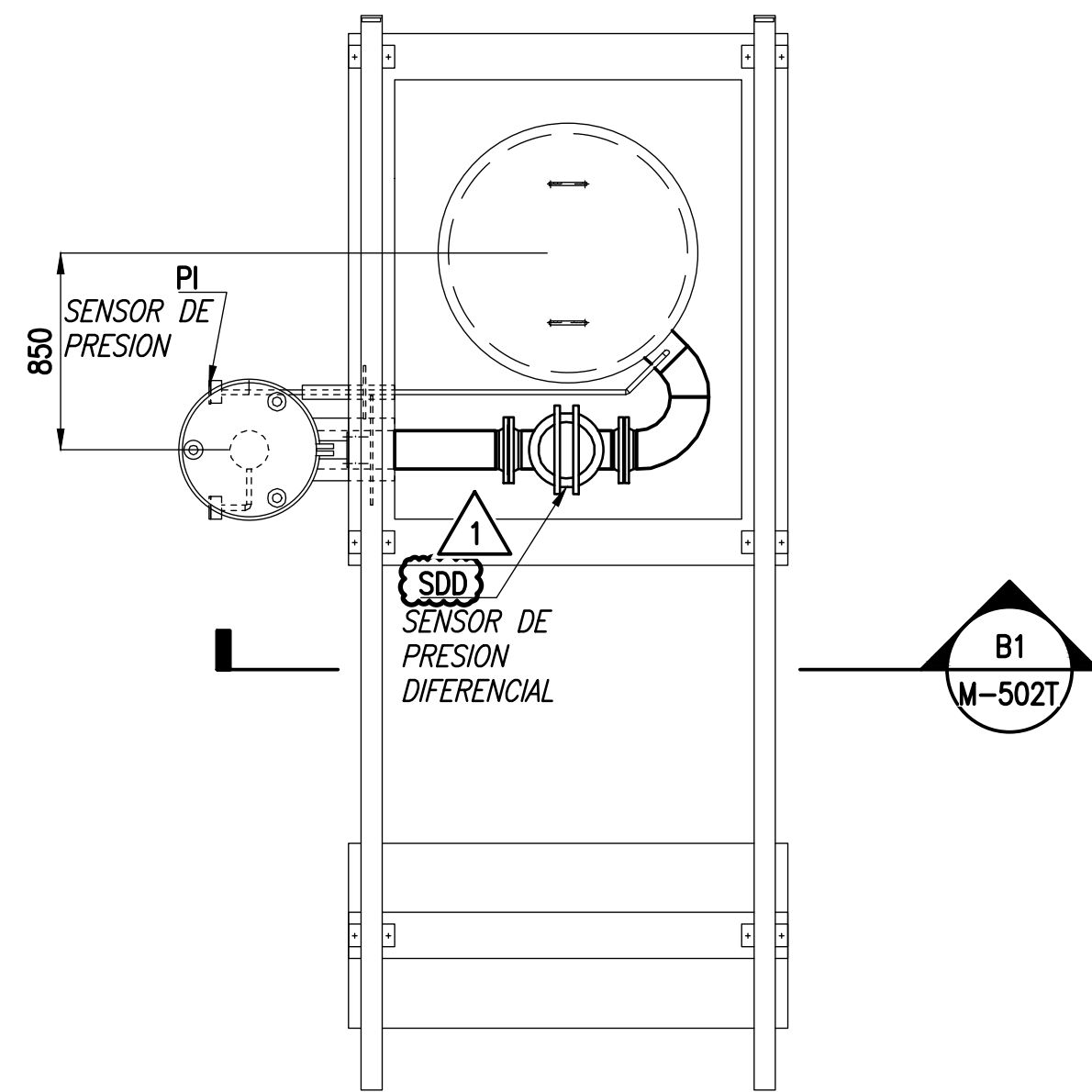
FILE NAME: P:\ABA\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - foto\CA0\WM-502T DETAILS.dwg LAYOUT NAME: M-502T DETAILS PLOTTED: Thursday, January 29, 2026 - 9:42am USER: rmgplns



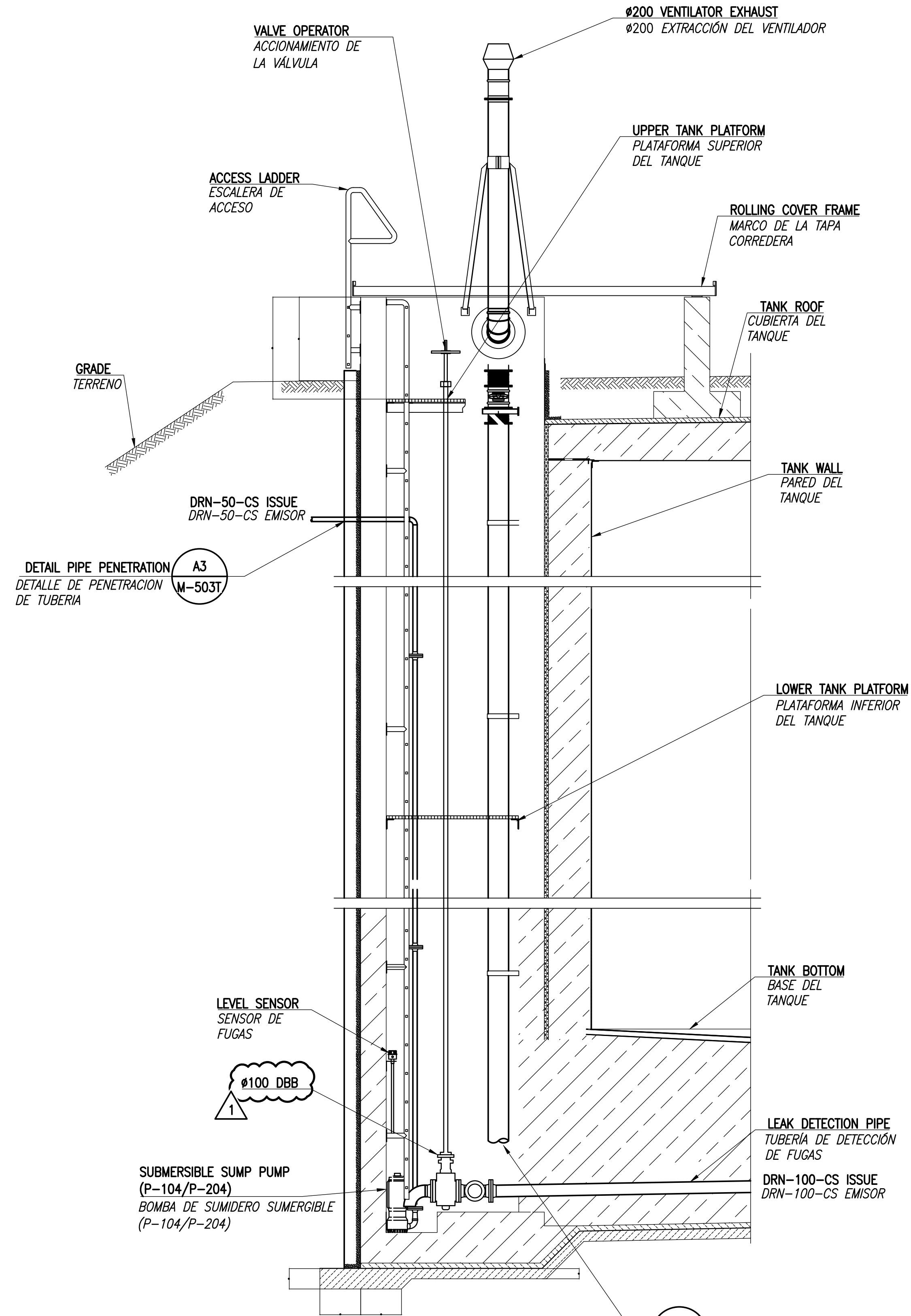
D1 LEAK DETECTION VAULT
CÁMARA DE DETECCIÓN DE FUGAS
SCALE: 1:30
ESCALA: 1:30



B1 SECTION
SECCIÓN
SCALE: NONE
ESCALA: TT



B1 MANHOLE WITH TANK VENT PIPING
ARQUETA CON TUBERÍA DE VENTEO DEL TANQUE
SCALE: 1:30
ESCALA: 1:30



A1 LEAK DETECTION WELL SECTION 1
POZO DE DETECCIÓN DE FUGAS
SECCIÓN 1
SCALE: 1:30
ESCALA: 1:30

NOTE: REFER TO M-201T FOR DRAIN PIPING DETAILS.
NOTAS: REFERIRSE A PLANO M-201T PARA DETALLES DE TUBERÍA DE DRENAJE

GRAPHIC SCALE(S):
ESCALA(S) GRÁFICA(S):

1:30 0 1000 2000 3000 mm



APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO: DATE

DES BMS DRW RTH CHK DWN

PM/SM CCA/ELA

BRANCH MANAGER

DESIGN MANAGER

FIRE PROTECTION

NAVAL FACILITIES ENGINEERING SYSTEM COMMAND

NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC

ATLANTIC DESIGN AND CONSTRUCTION

NAVAL STATION ROTA

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

ROTA, SPAIN

NORFOLK VA

SCALE:

PROJECT NO.:

CONSTR. CONTR. NO.:

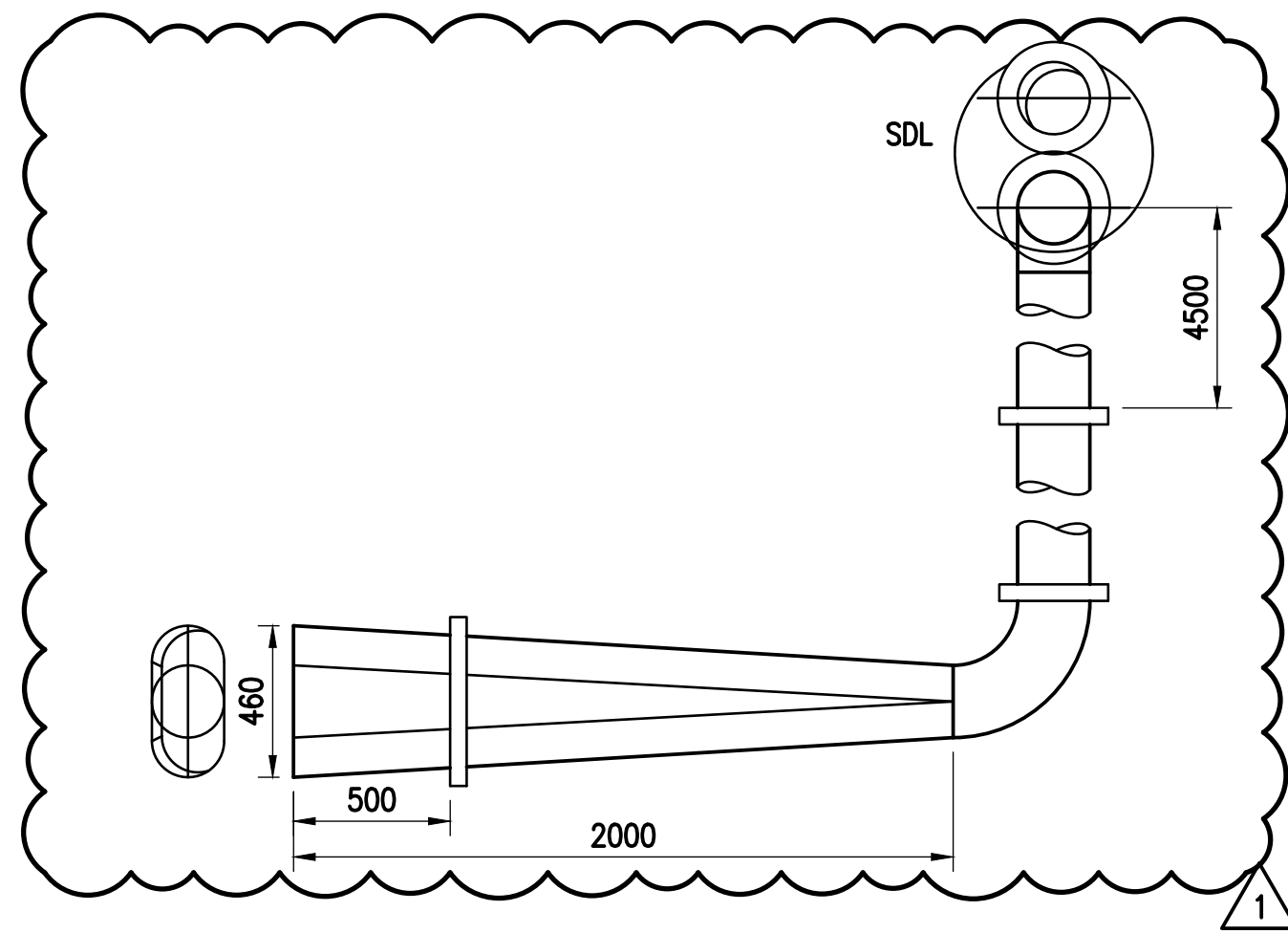
NAVFAC DRAWING NO. 14140188

SHEET 84 OF 342

M-502T

NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018

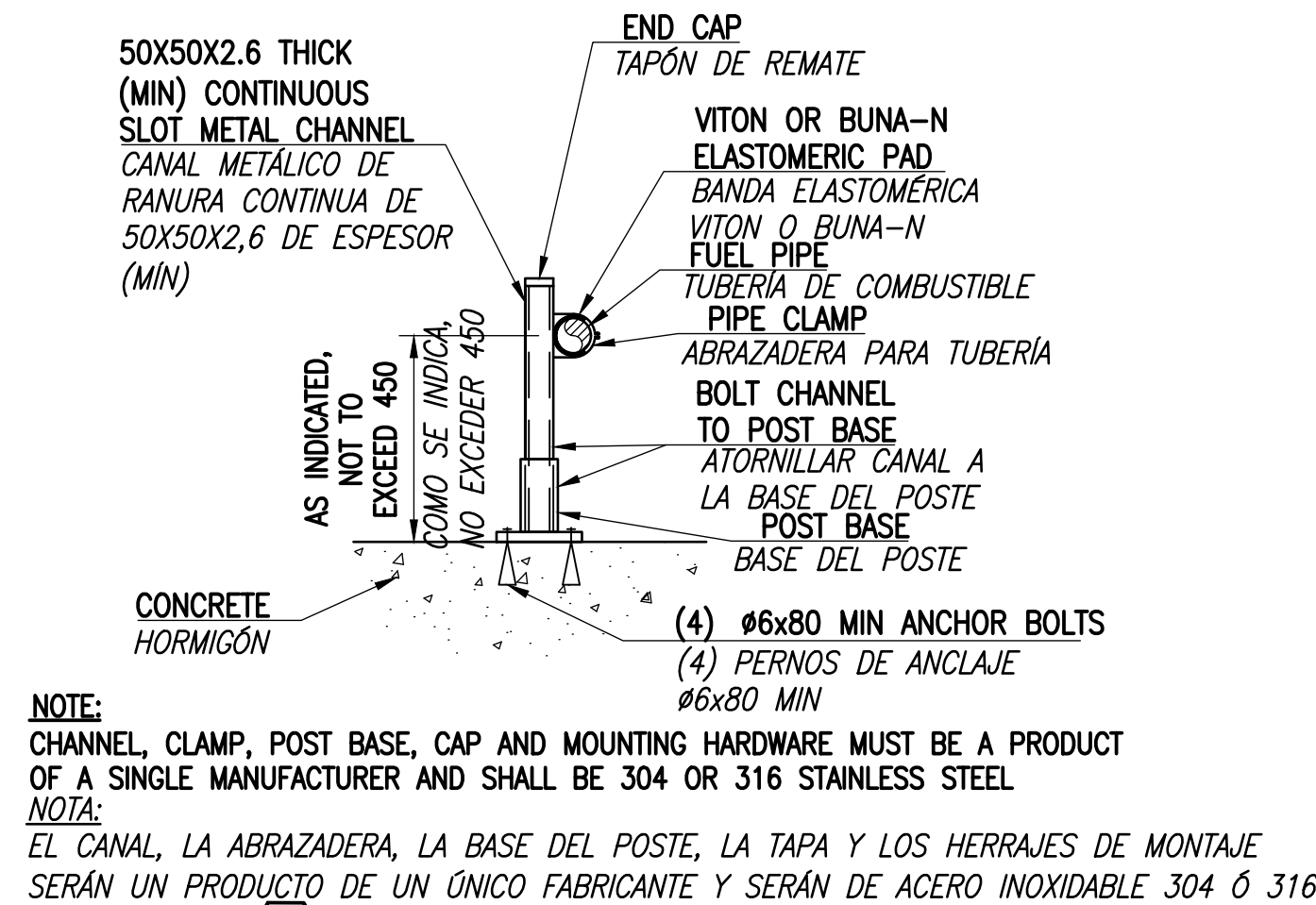
FILE NAME: P:\MBA\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Rota\CAO\WM-504T SUPPORT DETAILS.dwg LAYOUT NAME: M-504T PLOTTED: Thursday, January 29, 2026 - 9:43am USER: rhopkins



RECEIPT NOZZLE

BOQUILLA DE RECEPCIÓN

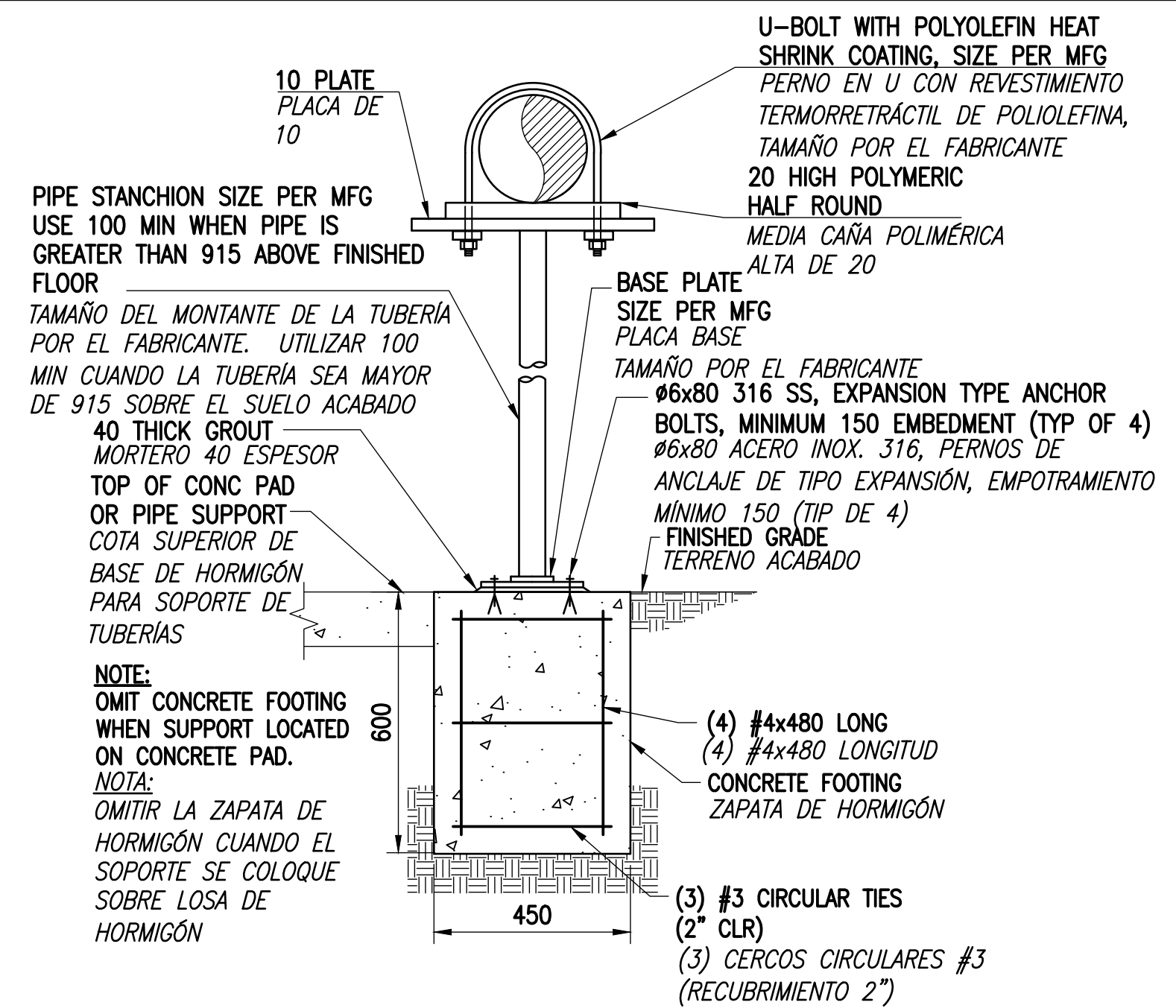
SCALE: NONE
ESCALA: SIN ESCALA



TYPE 1 SMALL PIPE SUPPORT

SOPORTE TUBERÍA PEQUEÑA TIPO 1

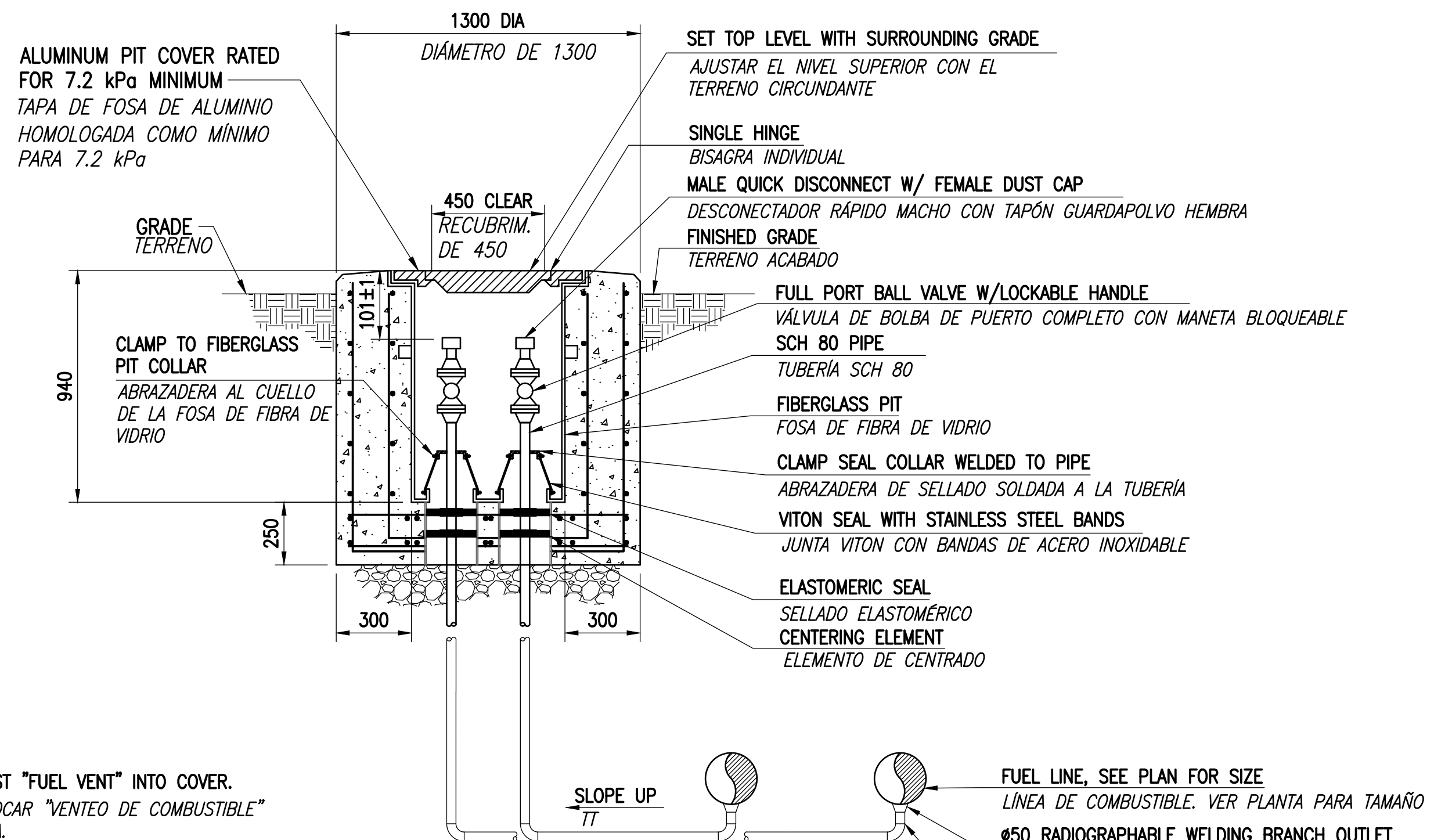
SCALE: NONE
ESCALA: SIN ESCALA



TYPE 2 FLOOR SUPPORTED LARGE PIPE SUPPORT

SOPORTE TUBERÍA LARGA APOYADO EN SUELO TIPO 2

SCALE: NONE
ESCALA: SIN ESCALA



LOW POINT DRAIN (LPD) PIT

FOSA DE DRENAJE EN PUNTO BAJO (LPD)

SCALE: NONE
ESCALA: SIN ESCALA



APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO: DATE

DESIGNER: BMS, DRAWN: RTH, CHECKED: DWN

PM/SM: CCA/ELA

BRANCH MANAGER

DESIGN MANAGER

FIRE PROTECTION

NAVAL FACILITIES ENGINEERING SYSTEM COMMAND

NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC

ATLANTIC DESIGN AND CONSTRUCTION

NAVAL STATION ROTA

ROTA, SPAIN

P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1

SUPPORT DETAILS

DETALLES DE SOPORTE

SCALE:

PROJECT NO.:

CONSTR. CONTR. NO.:

NAVFAC DRAWING NO.:

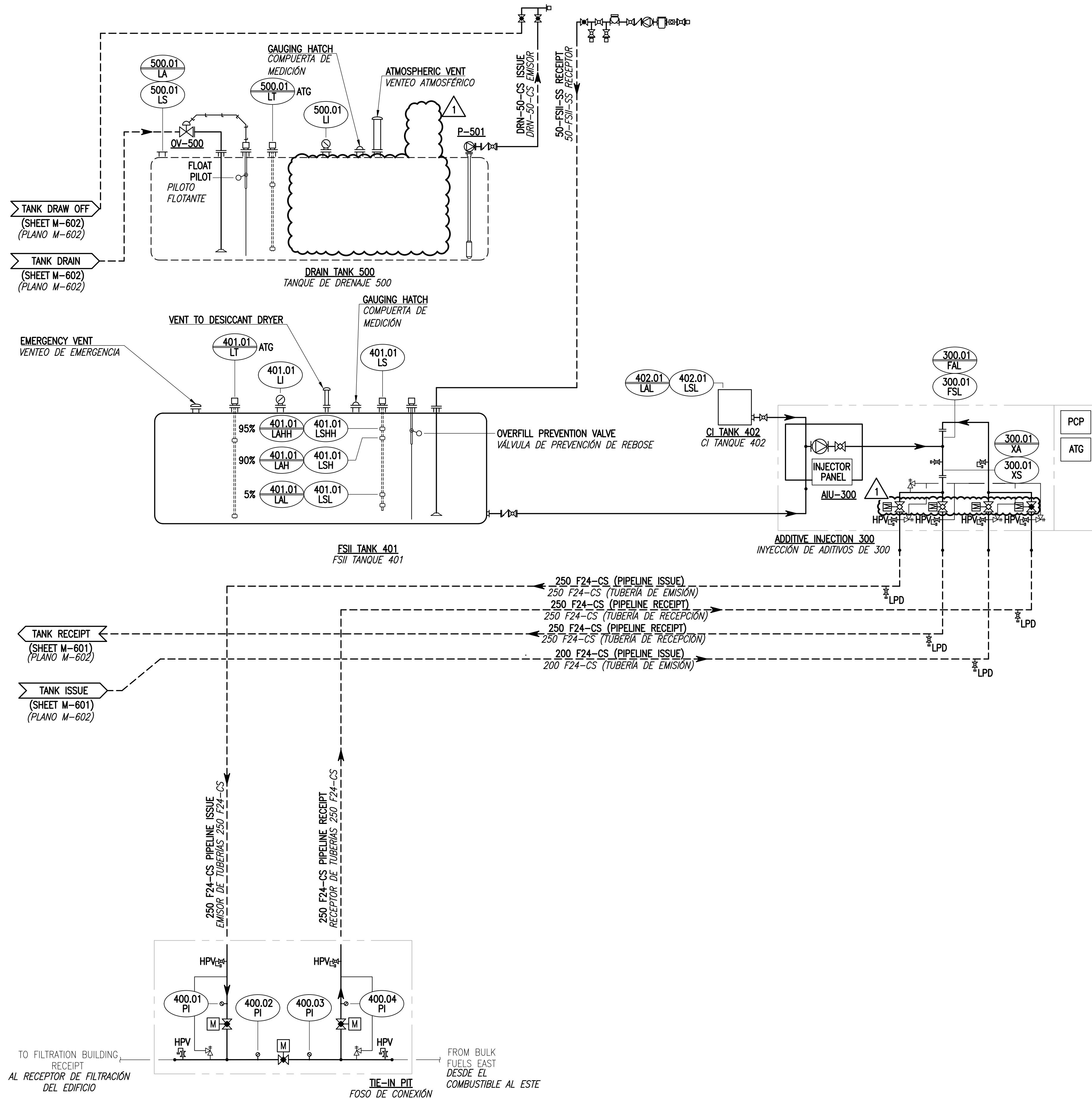
14140190

SHEET 86 OF 342

M-504T

NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018

FILE NAME: P:\NBA\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Roto\CAO\WM-602.dwg LAYOUT NAME: m-602 PLOTTED: Monday, February 23, 2026 - 8:49am USER: mapires



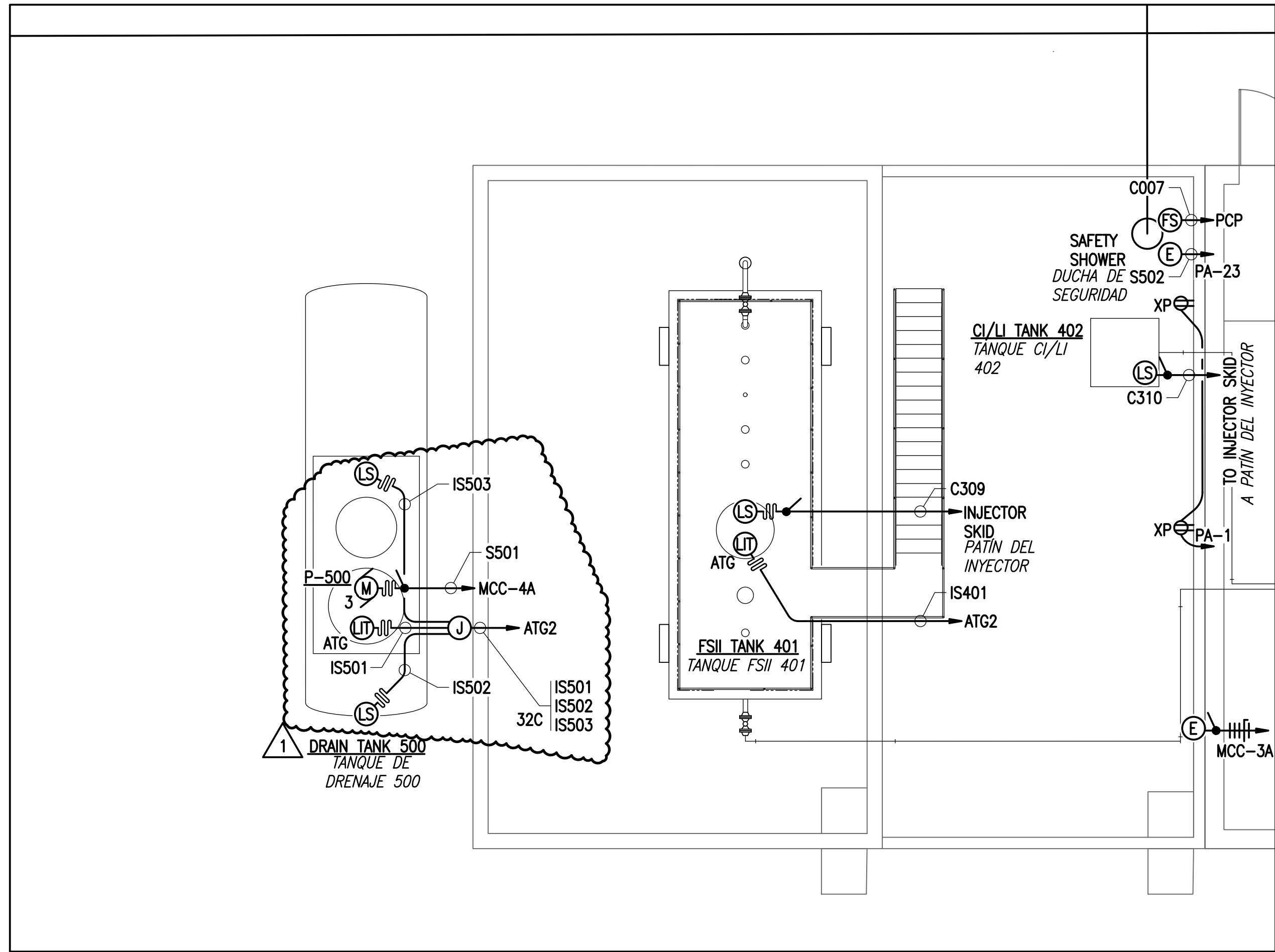
PIPING & INSTRUMENT DIAGRAM (ADDITIVE SYSTEM)
DIAGRAMA DE TUBERÍAS E INSTRUMENTACIÓN (SISTEMA DE ADITIVOS)
SCALE: NONE
ESCALA: SIN ESCALA

APPROVED	DATE	01/30/26
FOR COMMANDER NAVFAC	DATE	01/30/26
ACTIVITY	DATE	01/30/26
SATISFACTORY TO	DATE	01/30/26
DESIGNER	DATE	01/30/26
BRANCH MANAGER	DATE	01/30/26
DESIGN MANAGER	DATE	01/30/26
FIRE PROTECTION	DATE	01/30/26
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND	DATE	01/30/26
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC	DATE	01/30/26
NAVAL STATION - ROTA	DATE	01/30/26
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1	DATE	01/30/26
PIPING & INSTRUMENT DIAGRAM (ADDITIVE SYSTEM)	DATE	01/30/26
DIAGRAMA DE TUBERÍAS E INSTRUMENTACIÓN (SISTEMA DE ADITIVOS)	DATE	01/30/26
SCALE:	DATE	01/30/26
PROJECT NO.:	DATE	01/30/26
CONSTR. CONTR. NO.	DATE	01/30/26
NAVFAC DRAWING NO.	DATE	01/30/26
14140192	DATE	01/30/26
SHEET	DATE	01/30/26
88 OF 342	DATE	01/30/26
M-602T	DATE	01/30/26

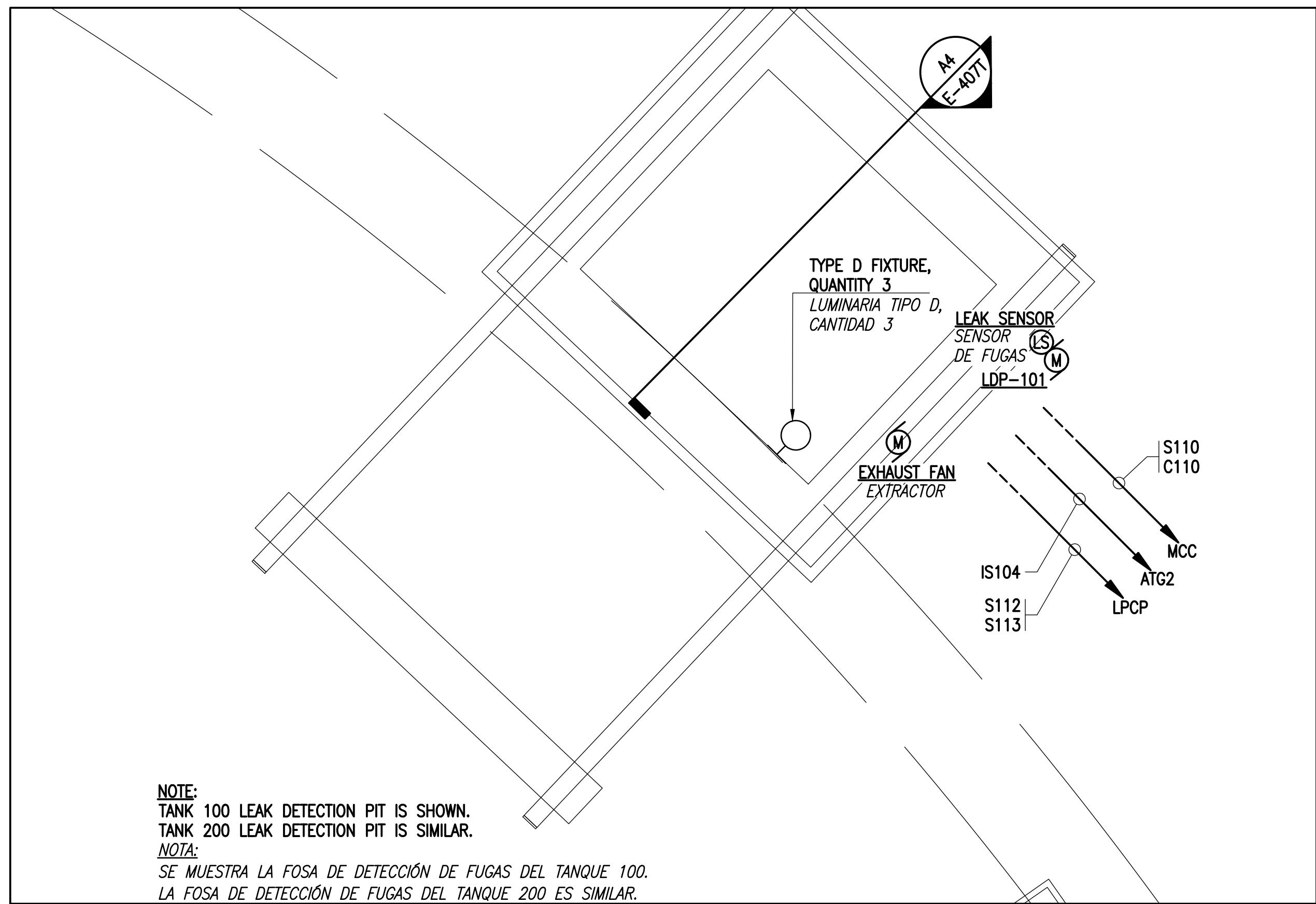
MARK	FLOW (L/S)	TDH (m)	RPM	EFFICIENCY	MOTOR DATA		SERVICE	CONFIGURATION	LOCATION	REMARKS
					DATOS DEL MOTOR					
MARCA	CAUDAL (L/S)			EFICIENCIA	KW (HP)	V/PH/Hz	SERVICIO	CONFIGURACIÓN	UBICACIÓN	OBSERVACIONES
P-101 P-102	38 L/S (600 GPM)	91 M	1800	70%	75 (100)	460/3ø/60	JET A-1 SURTIDOR 1-1	VERTICAL TURBINE TURBINA VERTICAL	TANK T-100 TANQUE T-100	MULTI-STAGE API 610 MULTI-ETAPA API 610
P-201 P-202	38 L/S (600 GPM)	91 M	1800	70%	75 (100)	460/3ø/60	JET A-1 SURTIDOR 1-1	VERTICAL TURBINE TURBINA VERTICAL	TANK T-200 TANQUE T-200	MULTI-STAGE API 610 MULTI-ETAPA API 610
P-103	3.15 L/S (50 GPM)	38 M	1800	70%	2 (3)	460/3ø/60	OILY WASTE RESIDUOS ACEITOSOS	VERTICAL TURBINE TURBINA VERTICAL	TANK T-100 TANQUE T-100	MULTI-STAGE API 610 MULTI-ETAPA API 610
P-104	0.75 L/S (10 GPM)	36 M	1800	70%	0.4 (0.5)	460/3ø/60	OILY WATER RESIDUOS ACEITOSOS	SUBMERSIBLE SUMERGIBLE	TANK T-100 TANQUE T-100	TANK T-100 TANQUE T-100
P-203	3.15 L/S (50 GPM)	38 M	1800	70%	2 (3)	460/3ø/60	OILY WASTE RESIDUOS ACEITOSOS	VERTICAL TURBINE TURBINA VERTICAL	TANK T-200 TANQUE T-200	MULTI-STAGE API 610 MULTI-ETAPA API 610
P-204	0.75 L/S (10 GPM)	36 M	1800	70%	0.4 (0.5)	460/3ø/60	OILY WATER RESIDUOS ACEITOSOS	SUBMERSIBLE SUMERGIBLE	TANK T-200 TANQUE T-200	TANK T-200 TANQUE T-200
P-501	3.15 L/S (50 GPM)	18 M	1800	70%	1 (3)	460/3ø/60	OILY WASTE RESIDUOS ACEITOSOS	VERTICAL TURBINE TURBINA VERTICAL	DRAIN TANK TANQUE DE DRENAJE	MULTI-STAGE API 610 MULTI-ETAPA API 610
P-601	10 L/S (150 GPM)	38 M	1800	70%	12 (15)	460/3ø/60	FSII FSII	POSITIVE DISPLACEMENT DESPLAZAMIENTO POSITIVO	FSII-OFF-LOAD FSII-DESCARGA	SELF PRIMING CENTRIFUGAL CENTRIFUGA AUTOCEBANTE

EQUIPMENT SCHEDULE				
CUADRO DE EQUIPOS				
EQUIPMENT NUMBER N° DE EQUIPO	EQUIPMENT NAME NOMBRE DEL EQUIPO	SERVICE SERVICIO	SIZE TAMAÑO	EQUIPMENT DESCRIPTION DESCRIPCIÓN DEL EQUIPO
ADDITIVE INJECTOR BUILDING				
AIU-300	INJECTION UNIT	VARIES	N/A	METERED PUMP INJECTION UNIT (3) ADDITIVES 38-76 L/S
	INYECTOR	VARIOS	N/A	INYECTOR CON BOMBA DOSIFICADORA (3) ADITIVOS 38-76 L/S
OV-500	OVERFILL VALVE VÁLVULA DE SOBRELLENADO	OILY WASTE RESIDUOS ACEITOSOS	50 50	FLOAT PILOT ACTUATED OVERFILL PROTECTION PROTECCIÓN CONTRA SOBRELLENADO ACCIONADA POR PILOTO FLOTANTE
P-501	WITHDRAWAL PUMP BOMBA DE EXTRACCIÓN	OILY WASTE RESIDUOS ACEITOSOS	50 (OUTLET) 50 (SALIDA)	SEE PUMP SCHEDULE VER CUADRO DE BOMBAS
T-401	ADDITIVE TANK TANQUE DE ADITIVOS	FSII FSII	N/A N/A	15,000 LITER (4,000 GALLONS) UL 142 AST 15.000 LITROS (4.000 GALONES) UL 142 AST
T-402	ADDITIVE TANK	CI	25 (OUTLET)	200 LITER (50 GAL) VESSEL SUPPLIED AS PART OF ADDITIVES INJECTION PACKAGE
	TANQUE DE ADITIVOS	CI	25 (SALIDA)	RECIPIENTE DE 1150 LITROS (300 GALONES) SUMINISTRADO COMO PARTE DEL PAQUETE DE INYECCIÓN DE ADITIVOS
T-500	DRAIN TANK TANQUE DE DRENAJE	OILY WASTE RESIDUOS ACEITOSOS	N/A N/A	15,000 LITER (4,000 GAL) UL 58 DOUBLE-WALL UST 15.000 LITROS (4.000 GALONES) UST DE DOBLE PARED
TANK 100				
FCV-101 FCV-102	NON-SURGE CHECK/PUMP CONTROL VALVE VÁLVULA ANTIRRETORNO/DE CONTROL DE LA BOMBA SIN SOBREPRESIÓN	JET A-1 SURTIDOR A-1	150	FLOW CONTROL = 38 L/s CHECK FEATURE FAIL CLOSED AGAINST REVERSE FLOW CONTROL DE CAUDAL = 38 L/S FUNCIÓN ANTIRRETORNO; FALLO CERRADO CONTRA FLUJO INVERSO
HLV-100	HIGH LIQUID LEVEL SHUT-OFF VALVE VÁLVULA DE CORTE PARA NIVEL DE LÍQUIDO ALTO	JET A-1 SURTIDOR A-1	200	FLOAT PILOT ACTUATED OVERFILL PROTECTION CHECK FEATURE; FAIL CLOSED AGAINST REVERSE FLOW PROTECCIÓN CONTRA SOBRELLENADO ACCIONADA POR PILOTO FLOTANTE; FUNCIÓN ANTIRRETORNO; FALLO CERRADO CONTRA FLUJO INVERSO
P-101 P-102	FUELING PUMP BOMBA DE COMBUSTIBLE	JET A-1 SURTIDOR A-1	150 (OUTLET) 150 (SALIDA)	SEE PUMP SCHEDULE VER CUADRO DE BOMBAS
P-103	WATER DRAW-OFF PUMP BOMBA DE EXTRACCIÓN DE AGUA	JET A-1 SURTIDOR A-1	50 (OUTLET) 50 (SALIDA)	SEE PUMP SCHEDULE VER CUADRO DE BOMBAS
P-104	LEAK DETECTION DRAIN PUMP BOMBA DE RESIDUOS ACEITOSOS	OILY WATER RESIDUOS ACEITOSOS	50 (OUTLET) 50 (SALIDA)	SEE PUMP SCHEDULE VER CUADRO DE BOMBAS
TANK 200				
FCV-201 FCV-202	NON-SURGE CHECK/PUMP CONTROL VALVE VÁLVULA ANTIRRETORNO/DE CONTROL DE LA BOMBA SIN SOBREPRESIÓN	JET A-1 SURTIDOR A-1	150	FLOW CONTROL = 38 L/s CHECK FEATURE FAIL CLOSED AGAINST REVERSE FLOW CONTROL DE CAUDAL = 38 L/S FUNCIÓN ANTIRRETORNO; FALLO CERRADO CONTRA FLUJO INVERSO
HLV-200	HIGH LIQUID LEVEL SHUT-OFF VALVE VÁLVULA DE CORTE PARA NIVEL DE LÍQUIDO ALTO	JET A-1 SURTIDOR A-1	200	FLOAT PILOT ACTUATED OVERFILL PROTECTION CHECK FEATURE; FAIL CLOSED AGAINST REVERSE FLOW PROTECCIÓN CONTRA SOBRELLENADO ACCIONADA POR PILOTO FLOTANTE; FUNCIÓN ANTIRRETORNO; FALLO CERRADO CONTRA FLUJO INVERSO
P-201 P-202	FUELING PUMP BOMBA DE COMBUSTIBLE	JET A-1 SURTIDOR A-1	150 (OUTLET) 150 (SALIDA)	SEE PUMP SCHEDULE VER CUADRO DE BOMBAS
P-203	WATER DRAW-OFF PUMP BOMBA DE EXTRACCIÓN DE AGUA	JET A-1 SURTIDOR A-1	50 (OUTLET)	SEE PUMP SCHEDULE
			50 (SALIDA)	VER CUADRO DE BOMBAS
P-204	LEAK DETECTION DRAIN PUMP BOMBA DE RESIDUOS ACEITOSOS	OILY WATER RESIDUOS ACEITOSOS	50 (OUTLET)	SEE PUMP SCHEDULE
			50 (SALIDA)	VER CUADRO DE BOMBAS
OFF-LOAD				
M-101	METER CONTADOR	FSII	80	
P-601	TRUCK OFF-LOAD PUMP BOMBA DE DESCARGA DE CAMIÓN	FSII	80 (OUTLET) 80 (SALIDA)	SEE PUMP SCHEDULE VER CUADRO DE BOMBAS

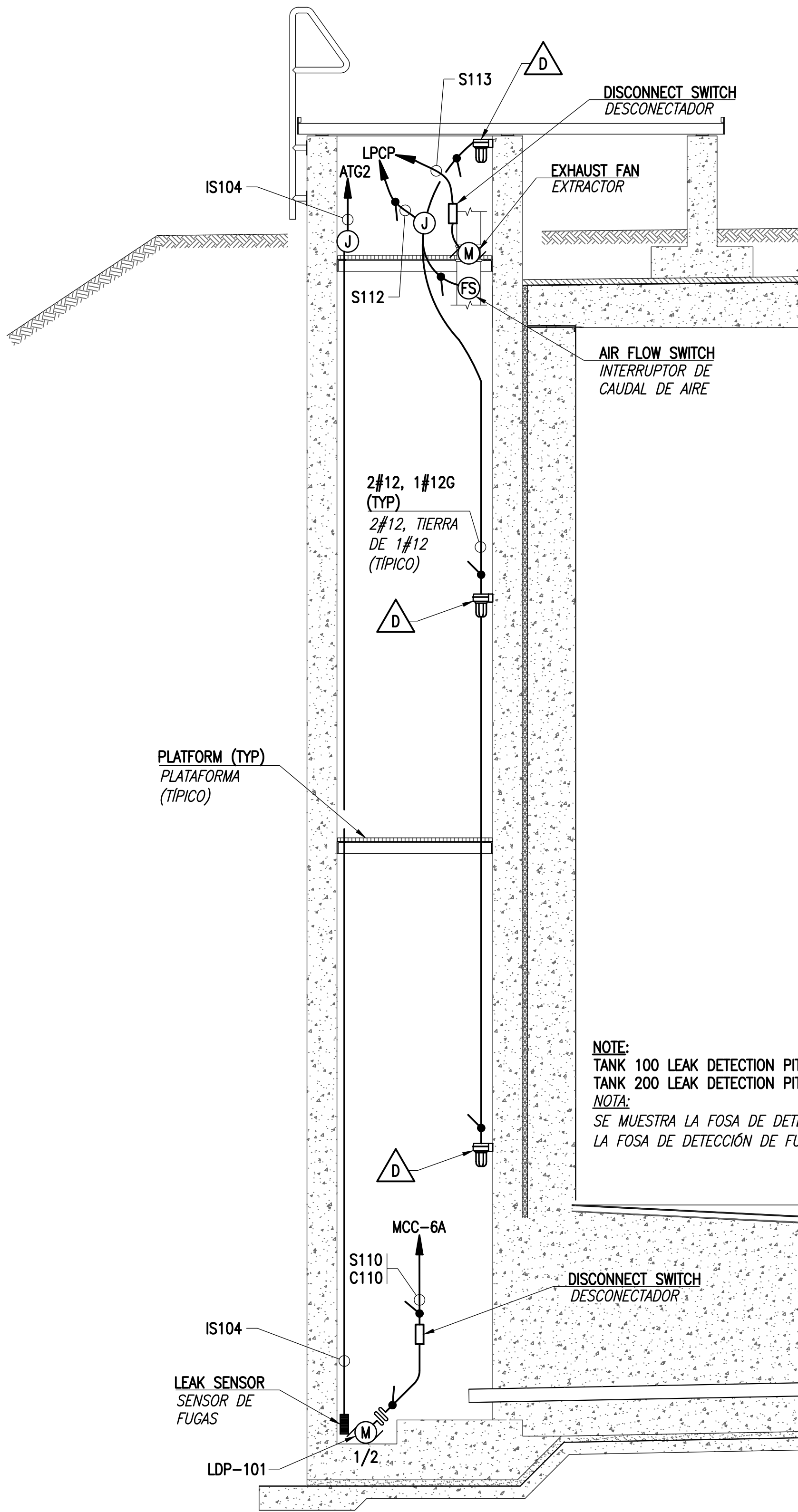
FILE NAME: P:\MBA\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Foto\CA0\E-407T ADDITIVE INJECTOR BUILDING.dwg LAYOUT NAME: E-407 PLOTTED: Thursday, January 29, 2026 - 9:43am USER: mopolina



ADDITIVE TANK AREA POWER AND CONTROLS PLAN
PLANTA DE FUERZA Y CONTROLES DE LA ZONA DEL TANQUE DE ADITIVOS
SCALE: 1:50
ESCALA: 1:50



LEAK DETECTION PIT ELECTRICAL PLAN
PLANTA ELÉCTRICA DE LA FOSA DE DETECCIÓN DE FUGAS
SCALE: 1:20
ESCALA: 1:20

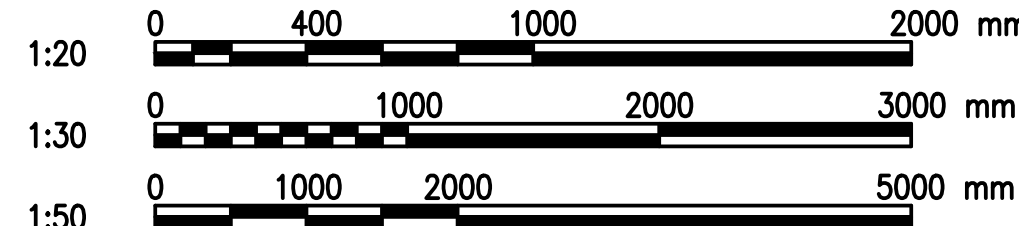


LEAK DETECTION WELL SECTION 1
SECCIÓN 1 DE POZO DE DETECCIÓN DE FUGAS
SCALE: 1:30
ESCALA: 1:30

GENERAL NOTES
NOTAS GENERALES

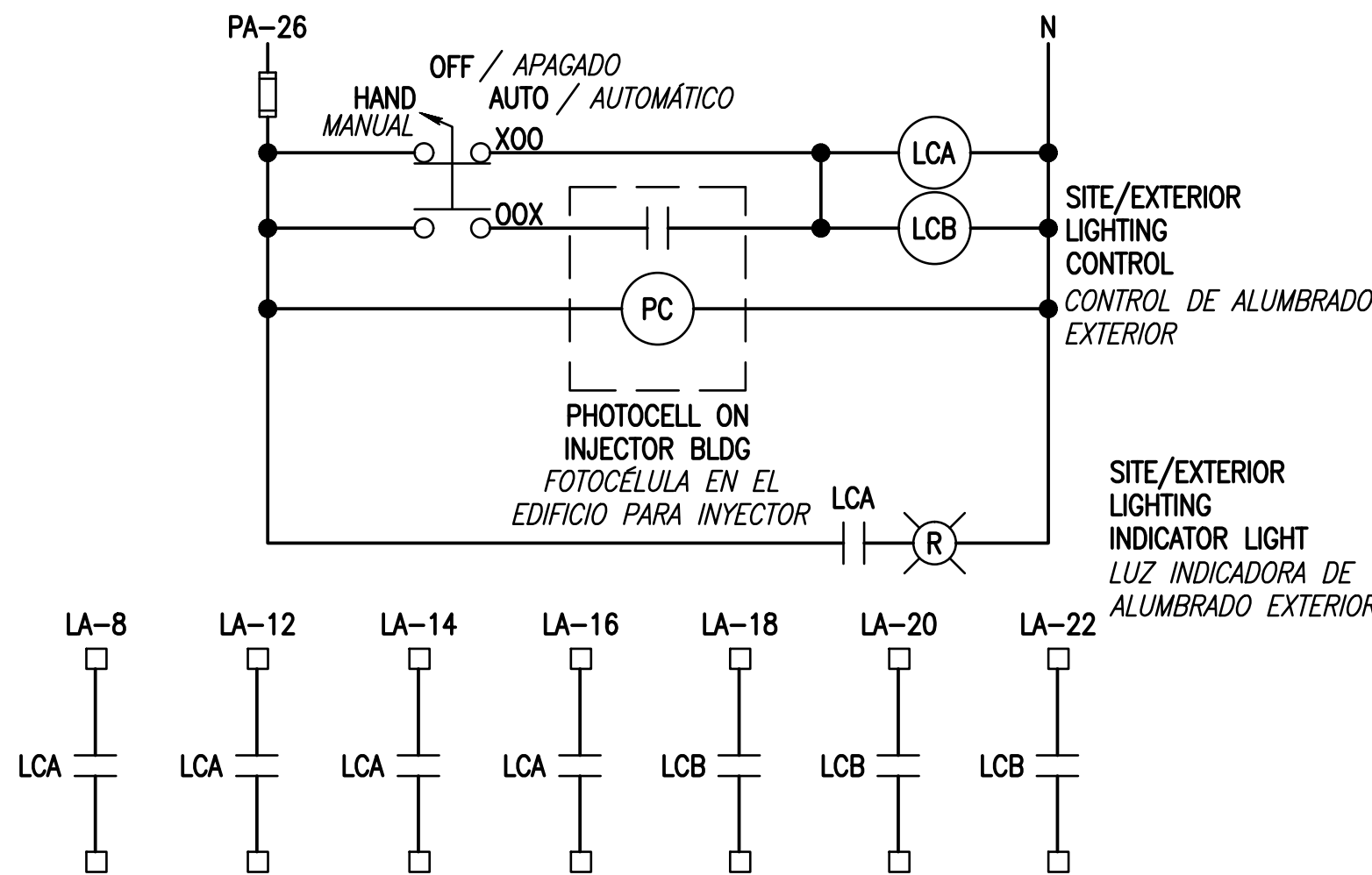
1. THE ADDITIVE TANK AREA, DRAIN TANK AREA, AND LEAK DETECTION PITS ARE HAZARDOUS LOCATIONS AS SHOWN ON E-705T AND E-706T. ELECTRICAL EQUIPMENT AND DEVICES INSTALLED WITHIN THE HAZARDOUS LOCATIONS MUST BE SPECIFICALLY APPROVED BY UL OR FACTORY MUTUAL FOR THE HAZARDOUS AREA CLASSIFICATION.
 2. EACH CONDUIT ORIGINATING IN OR PASSING THROUGH OR UNDER A HAZARDOUS AREA MUST HAVE EXPLOSIONPROOF SEALING FITTINGS INSTALLED IN THE INTERIOR OF THE PUMP VAULT PER NEC 501.15.
 3. WIRING METHODS WITHIN HAZARDOUS LOCATIONS MUST BE PROVIDED IN LISTED CLASS I LOCATION RMC CONDUIT, FLEXIBLE METALLIC CONDUIT, EXPLOSIONPROOF JUNCTION BOXES, AND CONDUIT SEALING FITTINGS UNLESS APPROVED OTHERWISE BY NEC ARTICLE 501.
 4. METALLIC CONDUITS THAT ARE NOT ATTACHED TO A GROUNDED PANEL OR ENCLOSURE MUST BE GROUNDED TO A GROUNDING BUSHING.
 5. WHERE A FLEXIBLE COUPLING IS USED, SUCH AS AT MOTOR TERMINALS, IT SHALL BE LISTED FOR A CLASS I, DIVISION 1 LOCATION WITH OPPOSITE END TERMINATED AT AN EXPLOSIONPROOF JUNCTION BOX THEN TO RIGID CONDUIT AND SEAL PRIOR TO TERMINATION AT CONTROL SWITCH OR AT CLASSIFICATION BOUNDARY.
1. LA ZONA DEL TANQUE DE ADITIVOS, LA ZONA DEL TANQUE DE DRENAJE Y LAS FOSAS DE DETECCIÓN DE FUGAS SON UBICACIONES PELIGROSAS COMO SE MUESTRA EN E-705T Y E-706T. LOS EQUIPOS Y DISPOSITIVOS ELÉCTRICOS INSTALADOS DENTRO DE LAS UBICACIONES PELIGROSAS DEBEN ESTAR APROBADOS ESPECÍFICAMENTE POR UL O FACTORY MUTUAL PARA LA CLASIFICACIÓN DE ZONA PELIGROSA.
 2. CADA CONDUCTO QUE TENGA SU ORIGEN EN UNA ZONA PELIGROSA, LA ATRAVIESE O PASE POR DEBAJO DE ELLA Y PENETRE EN LAS PAREDES DE LA CÁMARA DE BOMBAS DEBE TENER INSTALADOS ACCESORIOS DE SELLADO A PRUEBA DE EXPLOSIONES EN EL INTERIOR DE LA CÁMARA DE BOMBAS SEGÚN NEC 501.15.
 3. LOS MÉTODOS DE CABLEADO DENTRO DE LAS UBICACIONES PELIGROSAS DEBEN PROVEERSE EN CONDUCTOS RMC CLASIFICADOS PARA UBICACIÓN DE CLASE I, CONDUCTOS METÁLICOS FLEXIBLES, CAJAS DE CONEXIONES A PRUEBA DE EXPLOSIONES, Y ACCESORIOS DE SELLADO DE CONDUCTOS, A MENOS QUE SE APRUEBE LO CONTRARIO SEGÚN EL ARTÍCULO 501 DE NEC.
 4. LOS CONDUCTOS METÁLICOS QUE NO ESTÉN CONECTADOS A UN PANEL O RECINTO CON TOMA DE TIERRA DEBEN CONECTARSE A UN BORNE DE PUESTA A TIERRA.
 5. CUANDO SE UTILICE UN ACOPLAMIENTO FLEXIBLE, COMO EN LOS TERMINALES DEL MOTOR, SERÁ CLASIFICADO PARA UNA UBICACIÓN DE CLASE I, DIVISIÓN 1 CON EL EXTREMO OPUESTO TERMINADO EN UNA CAJA DE CONEXIONES A PRUEBA DE EXPLOSIONES Y LUEGO EN UN CONDUCTO RÍGIDO Y SELLAR ANTES DE LA TERMINACIÓN EN EL INTERRUPTOR DE CONTROL O EN EL LÍMITE DE LA CLASIFICACIÓN.

GRAPHIC SCALE(S):
ESCALA(S) GRÁFICA(S):



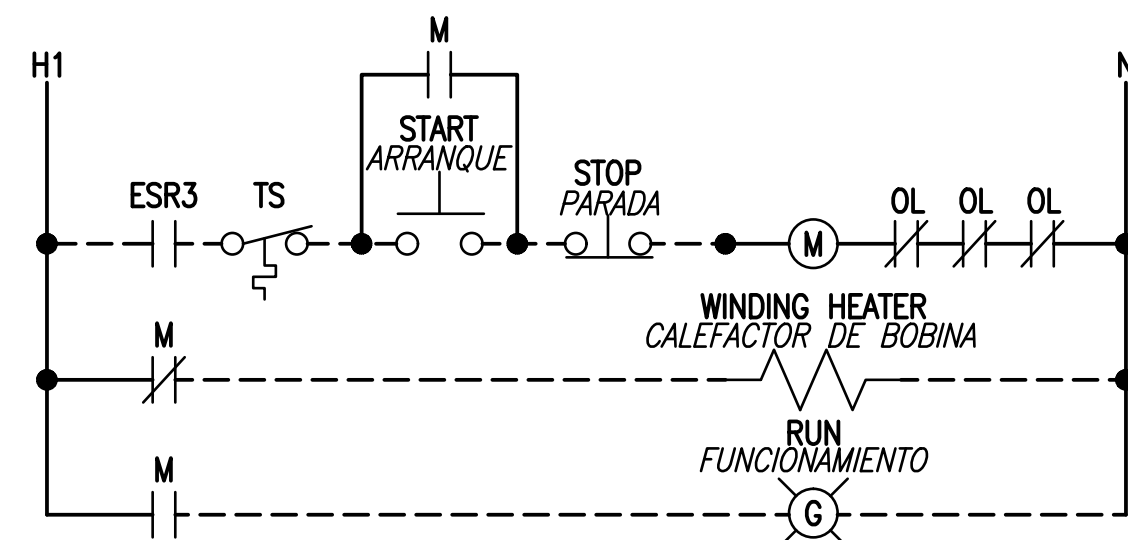
APPROVED		DATE	
FOR COMMANDER NAVFAC		10/30/26	
ACTIVITY		DRAIN TANK MODS	
SATISFACTORY TO		DATE	
DES	DST	DRW	RTH
PM/SM	CC/EA	CHK	CHM
BRANCH MANAGER		SEAL	
DESIGN MANAGER		A/E INFO	
FIRE PROTECTION		A	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND		NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC	
ATLANTIC DESIGN AND CONSTRUCTION		NORFOLK, VA	
NAVAL STATION ROTA		ROTA, SPAIN	
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1		ADDITIVE TANK AREA AND LEAK DETECTION PIT ELECTRICAL PLANS	
PL. ELECT. DE ZONA DEL TANQUE DE ADITIVOS Y DETECCIÓN DE FUGAS EN FOSA		E-407T	
SHEET 107 OF 342		NAVFAC DRAWING NO. 14140211	
E-407T		NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018	

FILE NAME: P:\MBA\21 Jobs\21-015 FY 23 MILCON P-919 Bulk Tanks - Foto(CAO)\E-6021 WIRING DIAGRAMS.dwg LAYOUT NAME: E-602 PLOTTED: Thursday, January 28, 2026 - 9:43am USER: nopolina



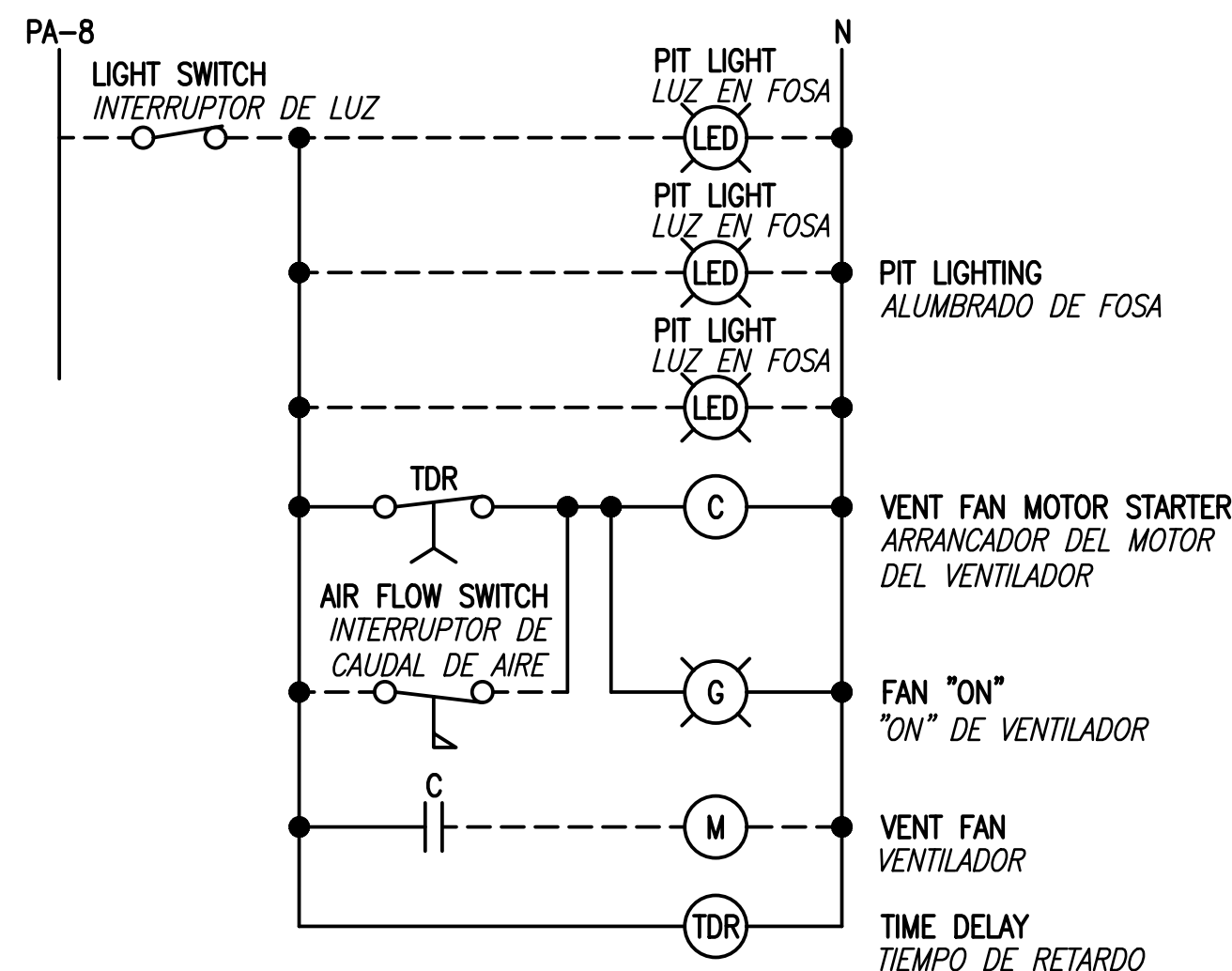
SITE/EXTERIOR LIGHTING CONTROL
CONTROL DE ALUMBRADO EXTERIOR

SCALE: NONE
ESCALA: SIN ESCALA



LEAK PIT PUMP WIRING DIAGRAM (LDP-101, LDP-201)
DIAGRAMA DE CABLEADO DE BOMBA PARA FUGAS EN FOSA (LDP-101, LDP-201)

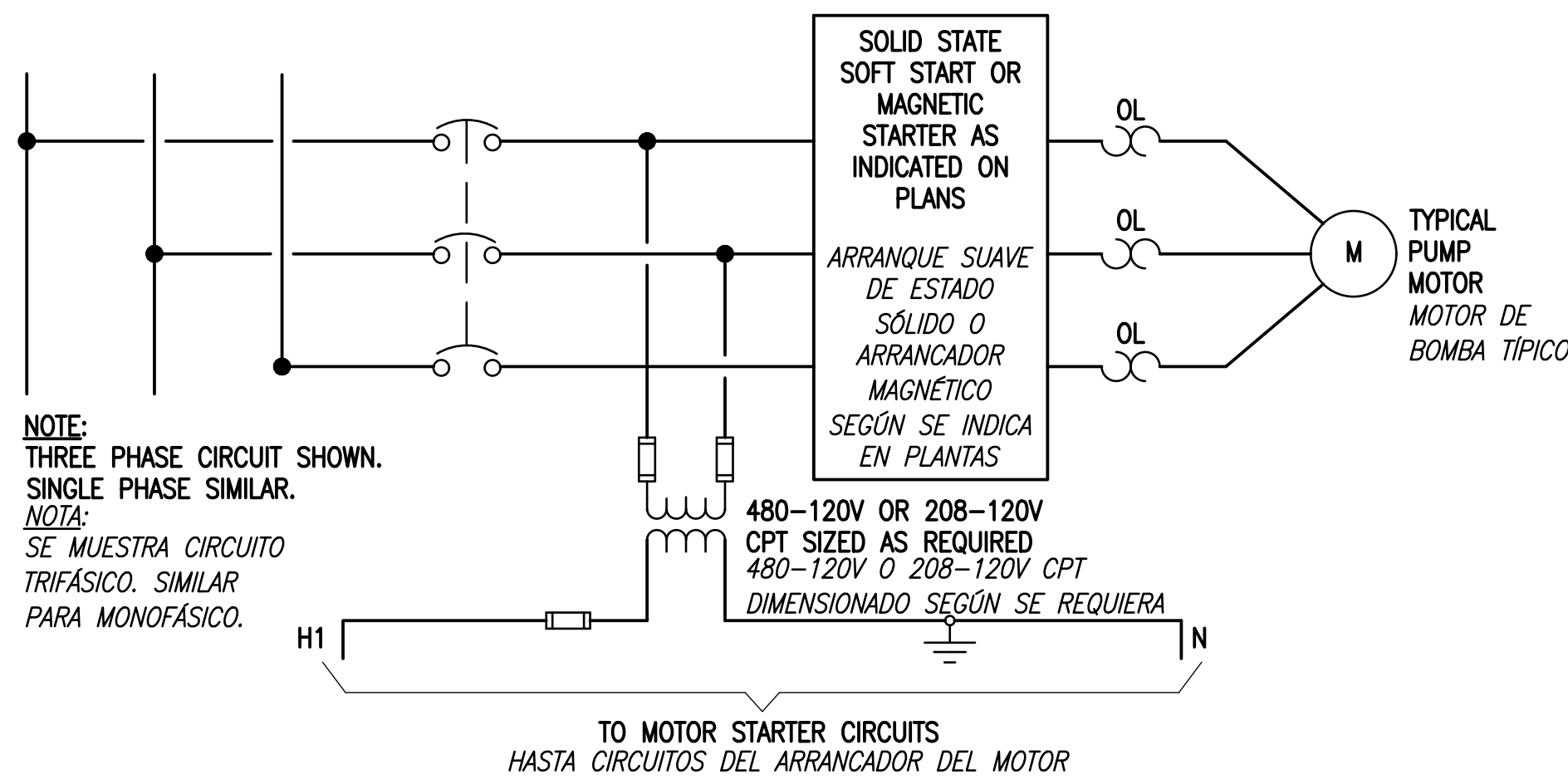
SCALE: NONE
ESCALA: SIN ESCALA



NOTE:
1. WIRING DIAGRAM FOR LPCP-1 IS SHOWN. WIRING FOR LPCP-2 IS SIMILAR.
NOTA:
1. SE MUESTRA EL DIAGRAMA DE CABLEADO PARA LPCP-1. EL CABLEADO PARA LPCP-2 ES SIMILAR.

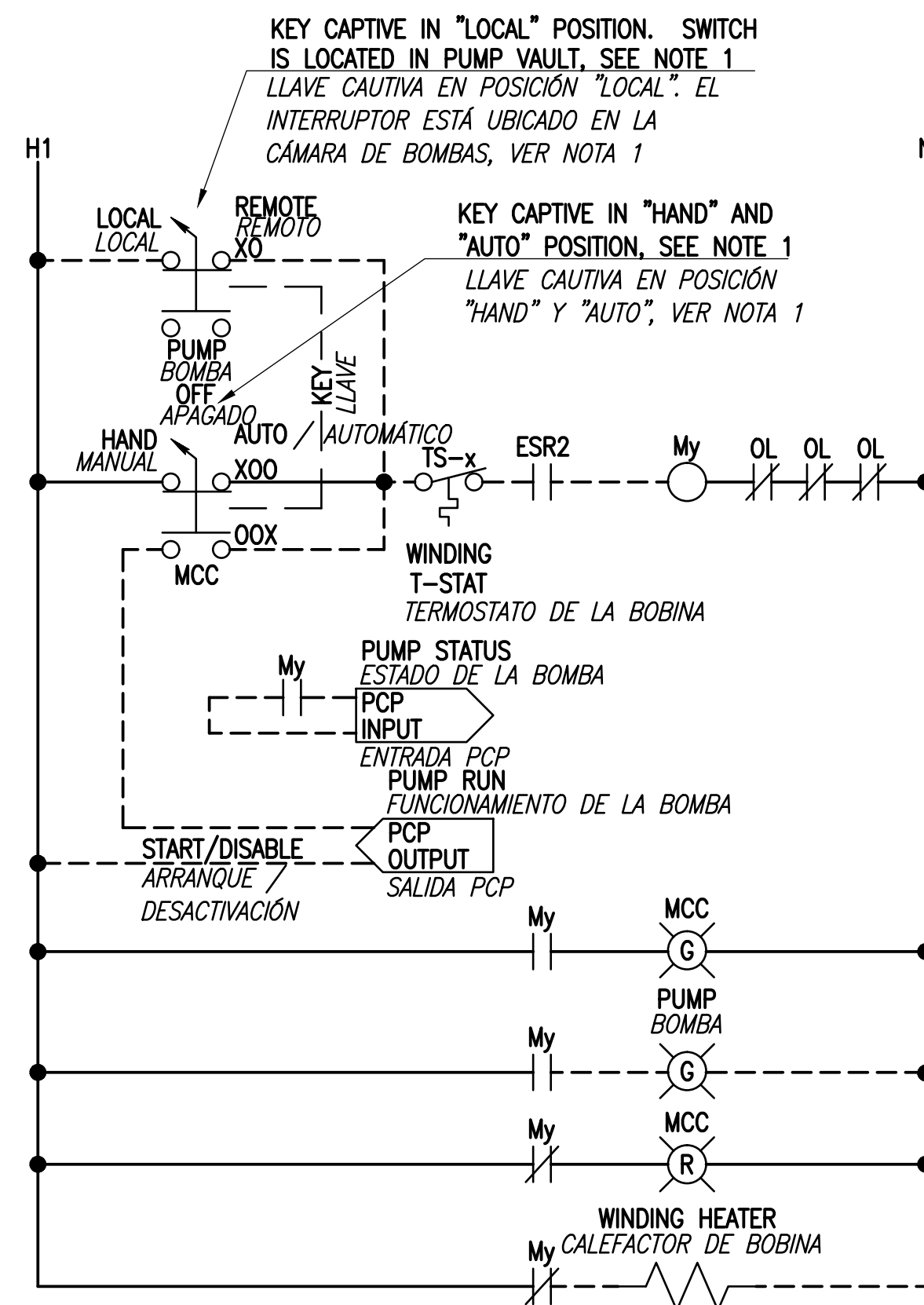
LEAK PIT CONTROL PANEL WIRING DIAGRAM
DIAGRAMA DE CABLEADO DEL PANEL DE CONTROL DE FUGAS EN FOSA

SCALE: NONE
ESCALA: SIN ESCALA



TYPICAL MOTOR STARTER CIRCUIT
CIRCUITO TÍPICO DEL ARRANCADOR DEL MOTOR

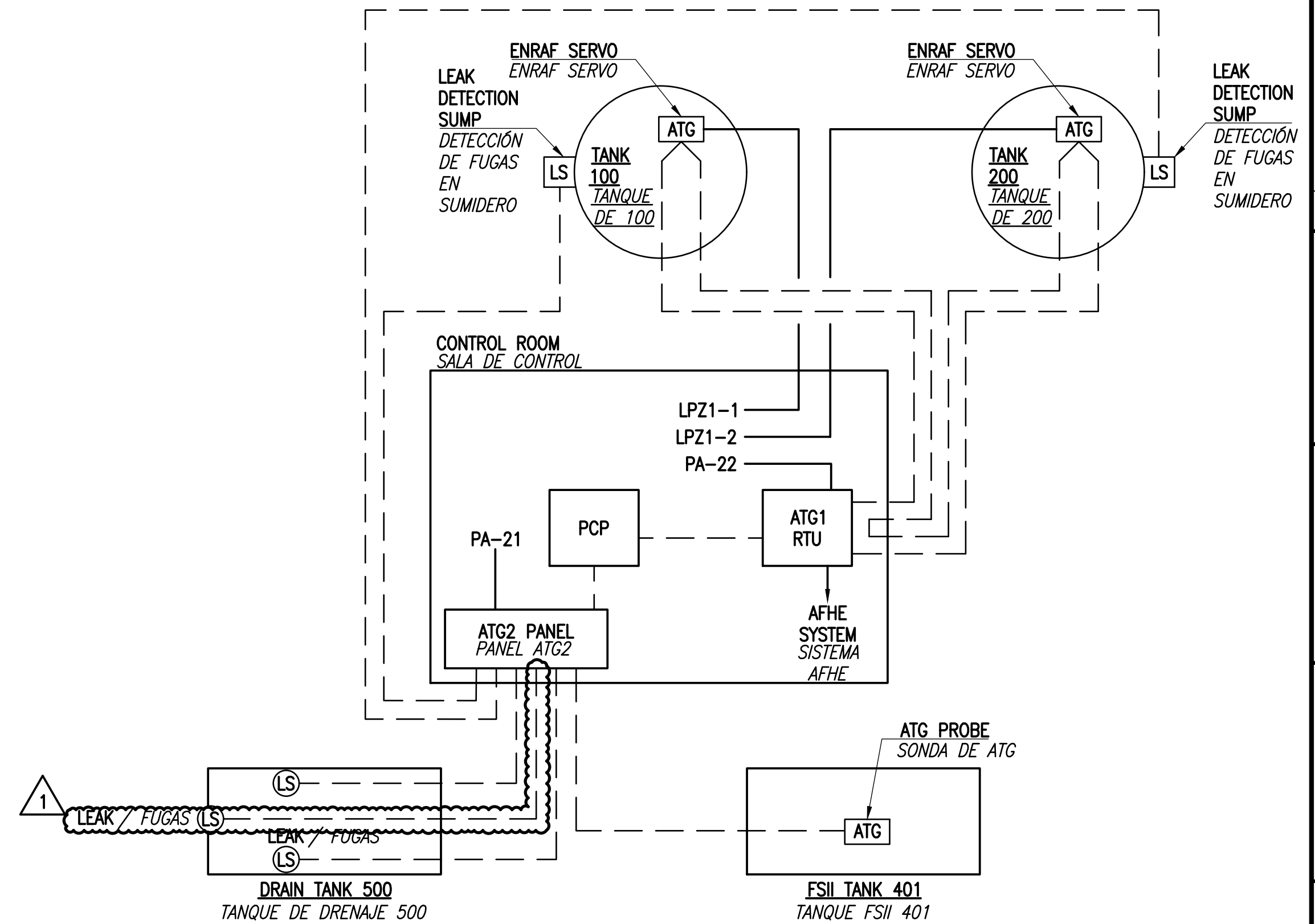
SCALE: NONE
ESCALA: SIN ESCALA



- NOTES: / NOTAS:
- KEYED LOCKS FOR LOCAL LOCAL-REMOTE SWITCHES AND CORRESPONDING MCC HOA SWITCHES SHALL BE KEYED ALIKE. KEYED LOCKS FOR LOCAL-REMOTE SWITCHES AND NON-CORRESPONDING MCC HOA SWITCHES SHALL NOT BE KEYED ALIKE.
 - TYPICAL FOR FUEL PUMPS P-101, P-102, P-103, P-201, P-202, AND P-203.
FOR PUMP P-101: x = 101, y = 1
FOR PUMP P-102: x = 102, y = 2
FOR PUMP P-103: x = 103, y = 3
FOR PUMP P-201: x = 201, y = 4
FOR PUMP P-202: x = 202, y = 5
FOR PUMP P-203: x = 203, y = 6
 - LAS CERRADURAS CON LLAVES PARA INTERRUPTORES LOCALES REMOTOS Y LOS INTERRUPTORES MCC HOA CORRESPONDIENTES TENDRÁN LA MISMA LLAVE. LAS CERRADURAS CON LLAVES PARA INTERRUPTORES LOCALES REMOTOS Y LOS INTERRUPTORES MCC HOA NO CORRESPONDIENTES NO TENDRÁN LLAVES IGUALES.
 - TÍPICO PARA BOMBAS PARA COMBUSTIBLE P-101, P-102, P-103, P-201, P-202, Y P-203.
PARA BOMBA P-101: x = 101, y = 1
PARA BOMBA P-102: x = 102, y = 2
PARA BOMBA P-103: x = 103, y = 3
PARA BOMBA P-201: x = 201, y = 4
PARA BOMBA P-202: x = 202, y = 5
PARA BOMBA P-203: x = 203, y = 6

FUEL PUMP WIRING DIAGRAM (P-x)
DIAGRAMA DE CABLEADO DE LA BOMBA PARA COMBUSTIBLE (P-x)

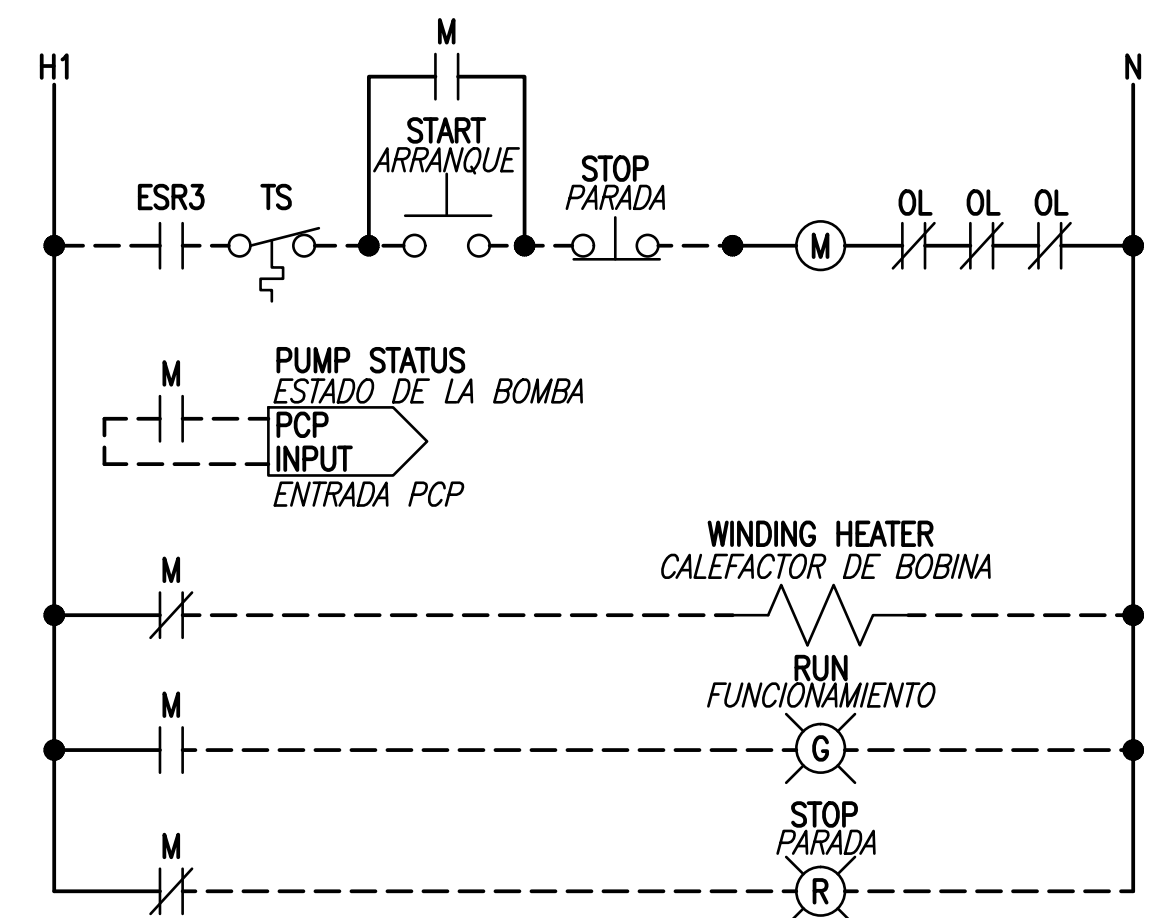
SCALE: NONE
ESCALA: SIN ESCALA



120V OR 240V POWER
FUERZA DE 120V O 240V
COMMUNICATIONS/CONTROL WIRING
CABLEADO DE COMUNICACIONES / CONTROL

ATG SYSTEM WIRING DIAGRAM
DIAGRAMA DE CABLEADO DEL SISTEMA ATG

SCALE: NONE
ESCALA: SIN ESCALA



DRAIN TANK PUMP P-501 WIRING DIAGRAM
DIAGRAMA DE CABLEADO DE LA BOMBA P-501 DEL TANQUE DE DRENAJE

SCALE: NONE
ESCALA: SIN ESCALA

1	2	3	4	5
DATE 01/30/26				
SYN DESCRIPTION 1 DRAIN TANK MODS				
APPR				
SEAL				
A/E INFO				
FOR COMMANDER NAVFAC				
ACTIVITY				
SATISFACTORY TO DATE				
DES DST DRW RTH CHK CHM				
PM/SM CCA/ELA				
BRANCH MANAGER				
DESIGN MANAGER				
FIRE PROTECTION				
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND				
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC				
NAVAL STATION ROTA				
ROTA, SPAIN				
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1				
WIRING DIAGRAMS				
DIAGRAMAS DE CABLEADO				
SCALE:				
PROJECT NO.:				
CONSTR. CONTR. NO.:				
NAVFAC DRAWING NO. 14140219				
SHEET 115 OF 342				
E-602T				
NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018				

NOTE:
1. PROVIDE MULTICONDUCTOR CONTROL CABLES AS TYPE TC. CABLES ROUTED EXTERIOR TO A BUILDING MUST BE TYPE TC-ER.

NOTA:
1. PROVEER CABLES DE CONTROL MULTICONDUCTORES DE TIPO TC. LOS CABLES CONDUCCIDOS EN EL EXTERIOR DE UN EDIFICIO DEBEN SER DE TIPO TC-ER.

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1. PROVIDE MULTICONDUCTOR CONTROL CABLES AS TYPE TC. CABLES ROUTED EXTERIOR TO A BUILDING MUST BE TYPE TC-ER.

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AVFAC METRIC DRAWFORM REVISION: 07 AUGUST 2018